
EBlink User Manual

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INTRODUCTION

EBlink is an ARM Cortex-M debug and flash programming tool, available for Windows, Linux x86_64, and Raspberry Pi (32/64-bit). It connects to the target through a hardware debug probe and is fully driven by Squirrel scripts, making it easy to add support for new devices without recompiling the tool.

1.1 Features

Flash programming

Erases, programs, and verifies target flash from ELF, Intel HEX, or SREC files. Only modified sectors are re-programmed by default. In-place byte patching, memory compare, and memory dump are also supported.

GDB server

Exposes a GDB remote-serial-protocol server so debuggers (arm-none-eabi-gdb, IDEs) can connect for source-level debugging. Supports up to four concurrent GDB connections per instance, flash caching for fast re-flash, and full semihosting. Multi-core targets (e.g. STM32H7x5) are handled with a separate GDB server per core.

Hotplug / non-intrusive inspection

Connect to a running target without causing a reset. Read and write memory, registers, and flash while the target is executing.

Script-driven target support

Every device-specific action — memory map, flash erase, flash write, custom reset — is implemented in a Squirrel script. Device scripts for Silicon Labs, STMicro, Atmel, NXP, and Renesas targets are included. Auto-detection is available via `-S auto`.

Exception unwinding

Fault stack frame unwinding with visual notification, configurable per target.

Script debugging

Breakpoints and variable watches in device scripts (since v6.0).

Real-time monitor (EBmon)

Standalone host-side monitor for non-GDB target control and background data logging (included since v6.21).

Remote debugging

GDB proxy support via EBridge for debugging across NAT networks or over Wi-Fi from a Raspberry Pi.

1.2 Supported probes

- **STlink V2 / V3** — including clones
- **CMSIS-DAP v1 / v2**

1.3 How it works

At startup EBLink:

1. Opens the debug probe specified with `-I`.
2. Loads the target script(s) specified with `-S`.
3. Calls `ebMain()` in the script, which sets up the memory map and initialises the device.
4. Carries out the requested operation — flash programming, GDB server, script execution, or a combination.

1.4 This manual

Usage

All command-line options, interface options, flash options, and GDB server options.

The script language

The Squirrel scripting language: syntax, types, control flow, and built-in functions.

EBlink Scripting

The EBlint scripting API: interface, device, target and buffer objects, error handling, constants, and the callback reference.

CHAPTER TWO

USAGE

This chapter walks you through using ELink from scratch — from connecting your first probe to running a GDB debug session.

2.1 Before you start

You need two things before EBlind can do anything:

1. A **debug probe** connected to your PC and to the target board (SWD or JTAG header).
2. A **device script** that tells EBlind about your target — its memory map, how to erase and write flash, and how to reset it.

EBlind is shipped with device scripts for STMicro, NXP, Atmel, Silicon Labs and Renesas targets. If you are not sure which script to use, start with `auto` and EBlind will identify the connected device and select the right script for you.

2.2 Your first command

The minimal EBlind command is:

```
eblink -I stlink -S auto
```

-I stlink tells EBlind to use an STlink probe. -S auto tells EBlind to auto-detect your device and load the right script.

If EBlind starts without errors, your probe is connected and the target is recognised. EBlind will then exit immediately because you have not asked it to do anything yet — that is normal.

2.3 Saving yourself some typing

Typing `-I stlink -S auto` on every command gets old fast. EBLink reads two environment variables at startup so you can set your defaults once and leave them out of every command:

EB_DEFAULT_PROBE

Used as the `-I` argument when `-I` is not on the command line.

EB_DEFAULT_SCRIPT

Used as a `-S` argument when `-S` is not on the command line.

Set them in your shell profile or system settings:

```
export EB_DEFAULT_PROBE=stlink
export EB_DEFAULT_SCRIPT=auto
```

After this, `eblink -F file=firmware.elf` is enough to flash a board — EBLink picks up the probe and script from the environment. Anything you do specify on the command line overrides the environment variable, so you can always deviate from the default for a single session without changing the setting:

```
# normally uses stlink from environment, override for this session:
eblink -I cmsis-dap -F file=firmware.elf
```

2.4 The command line structure

```
eblink -I <probe>[,options] -S <script> [operation] [options...]
```

EBlink always follows this sequence when it starts:

1. Open the debug probe (-I).
2. Load the device script(s) (-S).
3. Call `ebMain()` in the script — this sets up the memory map and initialises the target.
4. Carry out whatever operation you asked for (flash, GDB server, or both).

If any step fails, EBlink stops and reports the error. The most common first-time problems are a wrong probe type in -I, a missing USB driver, or a target that is not powered.

2.5 Configuring the probe: -I

The probe type comes first in the -I string, followed by comma-separated options:

```
-I stlink,speed=4000
```

The most useful options are:

speed=<kHz>

The SWD clock speed. The default is the maximum the probe supports, which is fine in most cases. If you get connection errors, try a lower speed such as `speed=1000` — long cables or noisy boards can cause problems at high speeds.

dr (disable reset)

Normally EBLink resets the target when it connects. Add `dr` to connect without a reset — useful when the target is already running code that must not be interrupted, or when the reset pin is not wired to the header.

swd / jtag

The debug protocol. SWD is the default and works on virtually all Cortex-M targets. Switch to `jtag` only if your hardware requires it.

serial=<sn>

If you have more than one probe plugged in at the same time, use the serial number to pick the right one. You can find serial numbers by running EBLink without this option — it will list all connected probes if there is ambiguity.

vcc=<V>

The target supply voltage (e.g. 3.3 or 1.8). EBLink uses this for speed calibration; it does not control the actual supply rail.

ap=<n>

By default EBLink scans the DAP and selects the first useful access port it finds. On multi-core devices such as the STM32H7, each core has its own AP, so you need to specify which one to use: `ap=0` for the CM7 core, `ap=1` for the CM4 core.

2.6 Loading a script: -S

The device script is what makes EBlind work with your specific target. It contains the memory map, the flash algorithm, and the reset strategy.

```
-S auto # auto-detect the device
-S stm32g0.script # explicit script by name
-S /home/user/scripts/my.script # explicit script by full path
```

With `auto`, EBlind reads the device ID register from the target and selects the matching script from the installed scripts. If the device is not recognised, EBlind tells you so and exits — in that case you need a custom script.

Use `-P <path>` to add a directory to the script search path so you can refer to scripts by filename rather than full path:

```
eblink -I stlink -S my_target.script -P /home/user/my_scripts
```

Use `-D name=value` to pass a value into the script as a global variable. Some scripts use this for configuration that differs between board variants:

```
-D FLASH_SIZE=0x2000000
```

2.7 Flashing firmware: -F

The -F option programs, verifies, or inspects flash memory. You combine keywords and values with commas:

```
-F file=firmware.elf,erase,verify,run
```

The most important thing to understand is that by default — without `erase` — EBLink only re-programs flash sectors that differ from the new image. Unchanged sectors are left alone. This is faster and reduces flash wear. It works for most development workflows.

Use `erase` when you need a clean slate: for example when changing partition layouts, clearing old data sectors, or recovering a board with write-protected flash.

The most common operations:

file=<file>

The image to program. EBLink accepts ELF, Intel HEX, and SREC and detects the format automatically from the file contents.

erase

Chip-erase the entire flash before programming. Use this when the default smart re-flash is not sufficient.

verify

Read back every programmed sector and compare it with the image. If anything does not match, EBLink reports an error. Adds some time but gives you certainty.

run

Release the target and let it start running after all operations are done. Without this the target stays halted.

cmp=<file>

Compare flash contents with a file without changing anything. Useful for checking whether a board already has the right firmware.

write=<hex>@<address>

Patch bytes at a specific flash address. The value is written as raw hex — two hex digits per byte, no `0x` prefix. The address uses the normal `0x` notation. This is done *after* the file upload, so you can patch a serial number or device-specific calibration into an otherwise identical firmware image:

```
-F file=fw.elf,write=0102030405@0x0800FF00
```

Multiple write options are allowed in a single -F.

read=<length>@<address>

Read bytes from the target and print them as hex. Handy for checking a register value or a specific memory location without a full debug session. Use `-v 0` to suppress all other EBLink output so only the hex bytes appear:

```
eblink -I stlink -S auto -v 0 -F read=4@0x08000004
```

dump=<length>@<address>:<file>

Save a region of memory to a file. Use a `.hex` extension for Intel HEX format, any other extension for raw binary.

Typical flashing examples:

```
# Development: only re-program what changed
eblink -I stlink -S auto -F file=firmware.elf

# Production: erase everything, program, verify, start
eblink -I stlink -S auto -F erase,verify,run,file=firmware.elf

# Check if the board already has the right firmware
eblink -I stlink -S auto -F cmp=firmware.elf
```

2.8 Debugging with GDB: -G

Adding -G starts a GDB remote-serial-protocol server. You then point your debugger (arm-none-eabi-gdb, an IDE like EmBitz or VS Code) at that server and connect:

```
eblink -I stlink -S auto -G
```

By default EBLink listens on all network interfaces on TCP port 2331. Your GDB client connects with:

```
target remote localhost:2331
```

The most useful -G options:

port=<n>

Use a different TCP port. Useful when the default port 2331 is already in use, or when you run multiple EBLink instances at the same time:

```
-G port=4242
```

ap=<n>

Attach this GDB server to a specific DAP access port. By default EBLink selects the first useful AP it finds. On a dual-core device (e.g. STM32H7), run two separate EBLink instances — one with -G ap=0 for the CM7 core and one with -G ap=1 for the CM4 core. Each GDB session then controls its own core independently.

s

Shut down EBLink when GDB disconnects. Without this, EBLink keeps running and waits for the next connection. Add s in automated test flows where EBLink should exit cleanly after each session.

nc

Disable the flash cache. EBLink normally caches flash contents so that re-flashing from GDB is fast. If your target modifies its own flash at runtime (bootloaders, EEPROM emulation), disable the cache so EBLink always reads back the real flash contents.

2.8.1 Combining -F and -G

You do not need to flash anything to use the GDB server — -G works on its own and connects to whatever is already running on the target. Adding -F is useful when you want to write something to flash first, for example a bootloader, a certificate, or device-specific calibration data, and then hand control to the debugger:

```
eblink -I stlink -S auto -F file=bootloader.elf -G port=4242
```

EBlink programs the flash first, then starts the GDB server and waits for a connection.

2.9 Configuring the Cortex-M core: -T

-T is optional and rarely needed for everyday use. EBLink auto-detects the core type. You add -T only when you want to change the default fault handling or reset behaviour.

fu=<0..2>

Fault unwind level. When your target crashes (HardFault, MemManage, etc.), EBLink can decode the exception stack frame and show you exactly which instruction faulted and why. 0 = disabled, 1 = passive (decode when halted), 2 = active (catch the fault immediately, default). This is one of EBLink's most useful debugging features — leave it at the default 2 unless you have a specific reason not to.

reset=<0..2>

Reset strategy: 0 = system reset (default), 1 = core-only reset, 2 = JTAG reset. Change this only if a system reset causes problems — for example when it reinitialises peripherals that need to stay powered.

fc=<flags>

Controls which fault classes trigger the vector catch. Default is all faults. Remove **r** if you do not want reset vectors caught:

```
-T cortex-m,fc=mncsbih
```

2.10 Other options

-v <0..8>

Verbosity level (default 4). Lower numbers give less output; higher numbers show internal debug traffic. Use **-v 0** for scripted use where you only want the result of a **-F read**.

-H

Connect without resetting the target — same as adding **dr** to the **-I** string.

-E <call>

Execute a Squirrel function after initialisation. Lets you call script functions from the command line without writing a wrapper script:

```
-E "dataflash_erase(0x08000500)"
```

Multiple **-E** options are executed in order.

-Q <n|y|d>

How Eblink handles yes/no questions from a script: **n** = always no, **y** = always yes, **d** = use the script default. Useful for unattended automation where no one is at the keyboard.

THE SCRIPT LANGUAGE

EBlink uses **Squirrel** as its scripting language — a high-level, imperative, object-oriented language designed by Alberto Demichelis as a lightweight yet powerful scripting tool. The language home page and full reference are at squirrel-lang.org.

Squirrel offers dynamic typing, delegation, higher-order functions, generators, tail recursion, exception handling, and automatic memory management, with both compiler and virtual machine fitting in about 6k lines of C++ code.

3.1 Lexical Structure

3.1.1 Identifiers

Identifiers start with an alphabetic character or the symbol ‘_’ followed by any number of alphabetic characters, ‘_’ or digits ([0-9]). Squirrel is a case sensitive language meaning that the lowercase and uppercase representation of the same alphabetic character are considered different characters. For instance, “foo”, “Foo” and “fOo” are treated as 3 distinct identifiers.

3.1.2 Keywords

The following words are reserved and cannot be used as identifiers:

base	break	case	catch	class	clone
continue	const	default	delete	else	enum
extends	for	foreach	function	if	in
local	null	resume	return	switch	this
throw	try	typeof	while	yield	constructor
instanceof	true	false	static	__LINE__	__FILE__

Keywords are covered in detail later in this document.

3.1.3 Operators

Squirrel recognizes the following operators:

!	!=		==	&&	>=	<=	>
<=>	+	+=	-	-=	/	/=	*
*=	%	%=	++	--	<-	=	&
^		~	>>	<<	>>>		

3.1.4 Other tokens

Other significant tokens are:

{	}	[	]	.	:
::	'	;	"	@	"

3.1.5 Literals

Squirrel accepts integer numbers, floating point numbers and string literals.

34	Integer number(base 10)
0xFF00A120	Integer number(base 16)
0753	Integer number(base 8)
'a'	Integer number
1.52	Floating point number
1.e2	Floating point number
1.e-2	Floating point number
"I'm a string"	String
@ "I'm a verbatim string"	String
@ " I'm a multiline verbatim string "	String

3.1.6 Comments

A comment is text that the compiler ignores but that is useful for programmers. Comments are normally used to embed annotations in the code. The compiler treats them as white space.

A comment can be `/*` (slash, asterisk) characters, followed by any sequence of characters (including new lines), followed by the `*/` characters. This syntax is the same as ANSI C.:

```
/*
this is
a multiline comment.
this lines will be ignored by the compiler
*/
```

A comment can also be `//` (two slashes) characters, followed by any sequence of characters. A new line not immediately preceded by a backslash terminates this form of comment. It is commonly called a “*single-line comment*.”:

```
//this is a single line comment. this line will be ignored by the compiler
```

The character `#` is an alternative syntax for single line comment.:

```
# this is also a single line comment.
```

This to facilitate the use of squirrel in UNIX-style shell scripts.

3.2 Values and Data types

While Squirrel is a dynamically typed language and variables do not have a type, different operations may interpret the variable as containing a type. Squirrel's basic types are integer, float, string, null, table, array, function, generator, class, instance, bool, thread and userdata.

3.2.1 Integer

An Integer represents a 32 bit (or better) signed number.:

```
local a = 123 //decimal
local b = 0x0012 //hexadecimal
local c = 075 //octal
local d = 'w' //char code
```

3.2.2 Float

A float represents a 32 bit (or better) floating point number.:

```
local a=1.0
local b=0.234
```

3.2.3 String

Strings are an immutable sequence of characters. In order to modify a string is it necessary create a new one.

Squirrel's strings are similar to strings in C or C++. They are delimited by quotation marks(") and can contain escape sequences (\t, \a, \b, \n, \r, \v, \f, \\, \", \', \0, \x<hh>, \u<hhhh> and \U<hhhhhhhh>).

Verbatim string literals do not interpret escape sequences. They begin with @" and end with the matching quote. Verbatim string literals also can extend over a line break. If they do, they include any white space characters between the quotes:

```
local a = "I'm a wonderful string\n"
// has a newline at the end of the string
local x = @"I'm a verbatim string\n"
// the \n is literal, similar to "\\n" in a regular string.
```

However, a doubled quotation mark within a verbatim string is replaced by a single quotation mark:

```
local multiline = @"
    this is a multiline string
    it will ""embed"" all the new line
    characters
"
```

3.2.4 Null

The null value is a primitive value that represents the null, empty, or non-existent reference. The type Null has exactly one value, called null.:

```
local a = null
```

3.2.5 Bool

Bool is a double-valued (Boolean) data type. Its literals are `true` and `false`. A bool value expresses the validity of a condition (tells whether the condition is true or false).:

```
local a = true;
```

3.2.6 Table

Tables are associative containers implemented as a set of key/value pairs called slots.:

```
local t={}
local test=
{
  a=10
  b=function(a) { return a+1; }
}
```

3.2.7 Array

Arrays are simple sequence of objects. Their size is dynamic and their index always starts from 0.:

```
local a = ["I'm", "an", "array"]
local b = [null]
b[0] = a[2];
```

3.2.8 Function

Functions are similar to those in other C-like languages with a few key differences (see below).

3.2.9 Class

Classes are associative containers implemented as sets of key/value pairs. Classes are created through a 'class expression' or a 'class statement'. class members can be inherited from another class object at creation time. After creation, members can be added until an instance of the class is created.

3.2.10 Class Instance

Class instances are created by calling a *class object*. Instances, as tables, are implemented as sets of key/value pairs. Instance members cannot be dynamically added or removed; however the value of the members can be changed.

3.2.11 Generator

Generators are functions that can be suspended with the statement ‘yield’ and resumed later (see [Generators](#)).

3.2.12 Userdata

Userdata objects are blobs of memory or pointers defined by the host application but stored within Squirrel variables.

3.2.13 Thread

Threads are objects representing a cooperative thread of execution, also known as coroutines.

3.2.14 Weak Reference

Weak References are objects that point to another (non-scalar) object but do not own a strong reference to it. (See [Weak References](#)).

3.3 Execution Context

The execution context is the union of the function stack frame and the function environment object(`this`) and the function root(`root table`). The stack frame is the portion of stack where the local variables declared in its body are stored. The environment object is an implicit parameter that is automatically passed by the function caller (see [functions](#)). The root table is a table associated to the function during its creation. The root table value of a function is the root table of the VM at the function creation. The root table of function can also be changed after creation with `closure.setroot()`. During the execution, the body of a function can only transparently refer to his execution context. This mean that a single identifier can refer to a local variable, to an environment object slot or to the slot of the closure root table; The environment object can be explicitly accessed by the keyword `this`. The closure root table can be explicitly accessed through the operator `::` (see [Variables](#)).

3.3.1 Variables

There are two types of variables in Squirrel, local variables and tables/arrays slots. Because global variables(variables stored in the root of a closure) are stored in a table, they are table slots.

A single identifier refers to a local variable or a slot in the environment object.:

```
derefexp := id;
```

```
_table["foo"]
_array[10]
```

with tables we can also use the `'.'` syntax:

```
derefexp := exp '.' id
```

```
_table.foo
```

Squirrel first checks if an identifier is a local variable (function arguments are local variables) if not looks up the environment object (`this`) and finally looks up to the closure root.

For instance::

```
function testy(arg)
{
    local a=10;
    print(a);
    return arg;
}
```

in this case `'foo'` will be equivalent to `'this.foo'` or `this["foo"]`.

Global variables are stored in a table called the root table. Usually in the global scope the environment object is the root table, but to explicitly access the closure root of the function from another scope, the slot name must be prefixed with `'::'` (`::foo`).

For instance::

```
function testy(arg)
{
    local a=10;
    return arg+::foo;
}
```

accesses the variable `'foo'` in the closure root table.

Since Squirrel 3.1 each function has a weak reference to a specific root table, this can differ from the current VM root table.:

```
function test() {  
    foo = 10;  
}
```

is equivalent to write:

```
function test() {  
    if("foo" in this) {  
        this.foo = 10;  
    }else {  
        ::foo = 10;  
    }  
}
```

3.4 Statements

A squirrel program is a simple sequence of statements.:

```
stats := stat [';' | '\n'] stats
```

Statements in squirrel are comparable to the C-Family languages (C/C++, Java, C# etc...): assignment, function calls, program flow control structures etc.. plus some custom statement like yield, table and array constructors (All those will be covered in detail later in this document). Statements can be separated with a new line or ';' (or with the keywords case or default if inside a switch/case statement), both symbols are not required if the statement is followed by '}'.

3.4.1 Block

```
stat := '{' stats '}'
```

A sequence of statements delimited by curly brackets ({ }) is called block; a block is a statement itself.

3.4.2 Control Flow Statements

squirrel implements the most common control flow statements: if, while, do-while, switch-case, for, foreach

true and false

Squirrel has a boolean type (bool) however like C++ it considers null, 0(integer) and 0.0(float) as *false*, any other value is considered *true*.

if/else statement

```
stat:= 'if' '(' exp ')' stat ['else' stat]
```

Conditionally execute a statement depending on the result of an expression.:

```
if(a>b)
    a=b;
else
    b=a;
////
if(a==10)
{
    b=a+b;
    return a;
}
```

while statement

```
stat := 'while' '(' exp ')' stat
```

Executes a statement while the condition is true.:

```
function testy(n)
{
    local a=0;
    while(a<n) a+=1;

    while(1)
    {
        if(a<0) break;
        a-=1;
    }
}
```

do/while statement

```
stat := 'do' stat 'while' '(' expression ')'
```

Executes a statement once, and then repeats execution of the statement until a condition expression evaluates to false.:

```
local a=0;
do
{
    print(a+"\n");
    a+=1;
} while(a>100)
```

switch statement

```
stat := 'switch' '(' exp ')' '{'
'case' case_exp ':'
    stats
['default' ':'
    stats]
'{'
```

Switch is a control statement allows multiple selections of code by passing control to one of the case statements within its body. The control is transferred to the case label whose case_exp matches with exp if none of the case match will jump to the default label (if present). A switch statement can contain any number if case instances, if 2 case have the same expression result the first one will be taken in account first. The default label is only allowed once and must be the last one. A break statement will jump outside the switch block.

3.4.3 Loops

for

```
stat := 'for' '(' [initexp] ';' [condexp] ';' [incexp] ')' statement
```

Executes a statement as long as a condition is different than false.:

```
for(local a=0;a<10;a+=1)
    print(a+"\n");
//or
glob <- null
for(glob=0;glob<10;glob+=1){
    print(glob+"\n");
}
//or
for(;;){
    print(loops forever+"\n");
}
```

foreach

```
'foreach' '(' [index_id',' value_id 'in' exp ']' stat
```

Executes a statement for every element contained in an array, table, class, string or generator. If exp is a generator it will be resumed every iteration as long as it is alive; the value will be the result of 'resume' and the index the sequence number of the iteration starting from 0.:

```
local a=[10,23,33,41,589,56]
foreach(idx,val in a)
    print("index="+idx+" value="+val+"\n");
//or
foreach(val in a)
    print("value="+val+"\n");
```

3.4.4 break

```
stat := 'break'
```

The break statement terminates the execution of a loop (for, foreach, while or do/while) or jumps out of switch statement;

3.4.5 continue

```
stat := 'continue'
```

The continue operator jumps to the next iteration of the loop skipping the execution of the following statements.

3.4.6 return

```
stat := return [exp]
```

The return statement terminates the execution of the current function/generator and optionally returns the result of an expression. If the expression is omitted the function will return null. If the return statement is used inside a generator, the generator will not be resumable anymore.

3.4.7 yield

```
stat := yield [exp]
```

(see *Generators*).

3.4.8 Local variables declaration

```
initz := id [= exp] [',' initz]  
stat := 'local' initz
```

Local variables can be declared at any point in the program; they exist between their declaration to the end of the block where they have been declared. *EXCEPTION*: a local declaration statement is allowed as first expression in a for loop.:

```
for(local a=0;a<10;a+=1)  
    print(a);
```

3.4.9 Function declaration

```
funcname := id ['::id]  
stat := 'function' id ['::id]+ '(' args ')' stat
```

creates a new function.

3.4.10 Class declaration

```
memberdecl := id '=' exp [';'] | '[' exp ']' '=' exp [';'] | functionstat |  
↳ 'constructor' functionexp  
stat := 'class' derefexp ['extends derefexp] '{'  
    [memberdecl]  
    '}'
```

creates a new class.

3.4.11 try/catch

```
stat := 'try' stat 'catch' '(' id ')' stat
```

The try statement encloses a block of code in which an exceptional condition can occur, such as a runtime error or a throw statement. The catch clause provides the exception-handling code. When a catch clause catches an exception, its id is bound to that exception.

3.4.12 throw

```
stat := 'throw' exp
```

Throws an exception. Any value can be thrown.

3.4.13 const

```
stat := 'const' id '=' 'Integer | Float | StringLiteral
```

Declares a constant (see *Constants & Enumerations*).

3.4.14 enum

```
enumerations := ( 'id' '=' Integer | Float | StringLiteral ) [',']  
stat := 'enum' id '{' enumerations '}'
```

Declares an enumeration (see *Constants & Enumerations*).

3.4.15 Expression statement

```
stat := exp
```

In Squirrel every expression is also allowed as statement, if so, the result of the expression is thrown away.

3.5 Expressions

3.5.1 Assignment

```
exp := derefexp '=' exp
exp:= derefexp '<-' exp
```

squirrel implements 2 kind of assignment: the normal assignment(=):

```
a = 10;
```

and the “new slot” assignment.:

```
a <- 10;
```

The new slot expression allows to add a new slot into a table(see [Tables](#)). If the slot already exists in the table it behaves like a normal assignment.

3.5.2 Operators

?: Operator

```
exp := exp_cond '?' exp1 ':' exp2
```

conditionally evaluate an expression depending on the result of an expression.

Arithmetic

```
exp:= 'exp' op 'exp'
```

Squirrel supports the standard arithmetic operators +, -, *, / and %. Other than that is also supports compact operators (+, -, *, /, %) and increment and decrement operators(++ and --):

```
a += 2;
//is the same as writing
a = a + 2;
x++
//is the same as writing
x = x + 1
```

All operators work normally with integers and floats; if one operand is an integer and one is a float the result of the expression will be float. The + operator has a special behavior with strings; if one of the operands is a string the operator + will try to convert the other operand to string as well and concatenate both together. For instances and tables, `_toString` is invoked.

Relational

```
exp := 'exp' op 'exp'
```

Relational operators in Squirrel are : ==, <, <=, >, >=, !=

These operators return true if the expression is false and a value different than true if the expression is true. Internally the VM uses the integer 1 as true but this could change in the future.

3 ways compare

```
exp := 'exp' <=> 'exp'
```

the 3 ways compare operator <=> compares 2 values A and B and returns an integer less than 0 if A < B, 0 if A == B and an integer greater than 0 if A > B.

Logical

```
exp := exp op exp
exp := '!' exp
```

Logical operators in Squirrel are : &&, ||, !

The operator && (logical and) returns null if its first argument is null, otherwise returns its second argument. The operator || (logical or) returns its first argument if is different than null, otherwise returns the second argument.

The '!' operator will return null if the given value to negate was different than null, or a value different than null if the given value was null.

in operator

```
exp := keyexp 'in' tableexp
```

Tests the existence of a slot in a table. Returns true if *keyexp* is a valid key in *tableexp*

```
local t=
{
    foo="I'm foo",
    [123]="I'm not foo"
}

if("foo" in t) dostuff("yep");
if(123 in t) dostuff();
```

instanceof operator

```
exp := instanceexp 'instanceof' classexp
```

Tests if a class instance is an instance of a certain class. Returns true if *instanceexp* is an instance of *classexp*.

typeof operator

```
exp:= 'typeof' exp
```

returns the type name of a value as string.:

```
local a={},b="squirrel"  
print(typeof a); //will print "table"  
print(typeof b); //will print "string"
```

Comma operator

```
exp:= exp ',' exp
```

The comma operator evaluates two expression left to right, the result of the operator is the result of the expression on the right; the result of the left expression is discarded.:

```
local j=0,k=0;  
for(local i=0; i<10; i++ , j++)  
{  
    k = i + j;  
}  
local a,k;  
a = (k=1,k+2); //a becomes 3
```

Bitwise Operators

```
exp:= 'exp' op 'exp'  
exp := '~' exp
```

Squirrel supports the standard C-like bitwise operators `&`, `|`, `^`, `~`, `<<`, `>>` plus the unsigned right shift operator `>>>`. The unsigned right shift works exactly like the normal right shift operator(`>>`) except for treating the left operand as an unsigned integer, so is not affected by the sign. Those operators only work on integer values; passing of any other operand type to these operators will cause an exception.

Operators precedence

<code>-</code> , <code>~</code> , <code>!</code> , <code>typeof</code> , <code>++</code> , <code>--</code>	highest
<code>/</code> , <code>*</code> , <code>%</code>	...
<code>+</code> , <code>-</code>	
<code><<</code> , <code>>></code> , <code>>>></code>	
<code><</code> , <code><=</code> , <code>></code> , <code>>=</code> , <code>instanceof</code>	
<code>==</code> , <code>!=</code> , <code><=></code>	
<code>&</code>	
<code>^</code>	
<code>&&</code> , <code>in</code>	
<code>+=</code> , <code>=</code> , <code>-=</code> , <code>/=</code> , <code>*=</code> , <code>%=</code>	...
<code>,</code> (comma operator)	lowest

3.5.3 Table Constructor

```
tslots := ( 'id' '=' exp | '[' exp ']' '=' exp ) [',']
exp := '{' [tslots] '}'
```

Creates a new table.:

```
local a = {} //create an empty table
```

A table constructor can also contain slots declaration; With the syntax:

```
local a = {
  slot1 = "I'm the slot value"
}
```

An alternative syntax can be:

```
'[' exp1 ']' = exp2 [',']
```

A new slot with exp1 as key and exp2 as value is created:

```
local a=
{
  [1]="I'm the value"
}
```

Both syntaxes can be mixed:

```
local table=
{
  a=10,
  b="string",
  [10]={},
  function bau(a,b)
  {
    return a+b;
  }
}
```

The comma between slots is optional.

Table with JSON syntax

Since Squirrel 3.0 is possible to declare a table using JSON syntax(see <http://www.wikipedia.org/wiki/JSON>). the following JSON snippet:

```
local x = {
  "id": 1,
  "name": "Foo",
  "price": 123,
  "tags": ["Bar", "Eek"]
}
```

is equivalent to the following squirrel code:

```
local x = {
  id = 1,
```

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```
name = "Foo",  
price = 123,  
tags = ["Bar", "Eek"]  
}
```

3.5.4 clone

```
exp := 'clone' exp
```

Clone performs shallow copy of a table, array or class instance (copies all slots in the new object without recursion). If the source table has a delegate, the same delegate will be assigned as delegate (not copied) to the new table (see *Delegation*).

After the new object is ready the “_cloned” meta method is called (see *Metamethods*).

When a class instance is cloned the constructor is not invoked (initializations must rely on ``_cloned`` instead

3.5.5 Array constructor

```
exp := '[' [explist] '']
```

Creates a new array.:

```
a <- [] //creates an empty array
```

Arrays can be initialized with values during the construction:

```
a <- [1, "string!", [], {}] //creates an array with 4 elements
```

3.6 Tables

Tables are associative containers implemented as pairs of key/value (called slot); values can be any possible type and keys any type except 'null'. Tables are squirrel's skeleton, delegation and many other features are all implemented through this type; even the environment, where "global" variables are stored, is a table (known as root table).

3.6.1 Construction

Tables are created through the table constructor

3.6.2 Slot creation

Adding a new slot in a existing table is done through the "new slot" operator <-; this operator behaves like a normal assignment except that if the slot does not exists it will be created.:

```
local a = {}
```

The following line will cause an exception because the slot named 'newslot' does not exist in the table 'a':

```
a.newslot = 1234
```

this will succeed:

```
a.newslot <- 1234;
```

or:

```
a[1] <- "I'm the value of the new slot";
```

3.6.3 Slot deletion

```
exp:= delete derefexp
```

Deletion of a slot is done through the keyword delete; the result of this expression will be the value of the deleted slot.:

```
a <- {
  test1=1234
  deleteme="now"
}

delete a.test1
print(delete a.deleteme); //this will print the string "now"
```

3.7 Arrays

An array is a sequence of values indexed by a integer number from 0 to the size of the array minus 1. Arrays elements can be obtained through their index.:

```
local a=["I'm a string", 123]
print(typeof a[0]) //prints "string"
print(typeof a[1]) //prints "integer"
```

Resizing, insertion, deletion of arrays and arrays elements is done through a set of standard functions (see *built-in functions*).

3.8 Functions

Functions are first class values like integer or strings and can be stored in table slots, local variables, arrays and passed as function parameters. Functions can be implemented in Squirrel or in a native language with calling conventions compatible with ANSI C.

3.8.1 Function declaration

Functions are declared through the function expression:

```
local a = function(a, b, c) { return a + b - c; }
```

or with the syntactic sugar:

```
function ciao(a,b,c)
{
    return a+b-c;
}
```

that is equivalent to:

```
this.ciao <- function(a,b,c)
{
    return a+b-c;
}
```

a local function can be declared with this syntactic sugar:

```
local function tuna(a,b,c)
{
    return a+b-c;
}
```

that is equivalent to:

```
local tuna = function(a,b,c)
{
    return a+b-c;
}
```

is also possible to declare something like:

```
T <- {}
function T::ciao(a,b,c)
{
    return a+b-c;
}

//that is equivalent to write

T.ciao <- function(a,b,c)
{
    return a+b-c;
}

//or
```

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```
T <- {  
  function ciao(a,b,c)  
  {  
    return a+b-c;  
  }  
}
```

Default Paramaters

Squirrel's functions can have default parameters.

A function with default parameters is declared as follows:

```
function test(a,b,c = 10, d = 20)  
{  
  . . . .  
}
```

when the function *test* is invoked and the parameter *c* or *d* are not specified, the VM automatically assigns the default value to the unspecified parameter. A default parameter can be any valid squirrel expression. The expression is evaluated at runtime.

Function with variable number of paramaters

Squirrel's functions can have variable number of parameters(varargs functions).

A vararg function is declared by adding three dots (...) at the end of its parameter list.

When the function is called all the extra parameters will be accessible through the *array* called *vargv*, that is passed as implicit parameter.

vargv is a regular squirrel array and can be used accordingly.:

```
function test(a,b,...)  
{  
  for(local i = 0; i< vargv.len(); i++)  
  {  
    ::print("varparam "+i+" = "+vargv[i]+"\\n");  
  }  
  foreach(i,val in vargv)  
  {  
    ::print("varparam "+i+" = "+val+"\\n");  
  }  
}  
  
test("goes in a","goes in b",0,1,2,3,4,5,6,7,8);
```

3.8.2 Function calls

```
exp := derefexp '(' explist ')'
```

The expression is evaluated in this order: derefexp after the explist (arguments) and at the end the call.

A function call in Squirrel passes the current environment object *this* as a hidden parameter. But when the function was immediately indexed from an object, *this* shall be the object which was indexed, instead.

If we call a function with the syntax:

```
mytable.foo(x,y)
```

the environment object passed to 'foo' as *this* will be 'mytable' (since 'foo' was immediately indexed from 'mytable')

Whereas with the syntax:

```
foo(x,y) // implicitly equivalent to this.foo(x,y)
```

the environment object will be the current *this* (that is, propagated from the caller's *this*).

It may help to remember the rules in the following way:

foo(x,y) —> this.foo(x,y) table.foo(x,y) —> call foo with (table,x,y)

It may also help to consider why it works this way: it's designed to assist with object-oriented style. When calling 'foo(x,y)' it's assumed you're calling another member of the object (or of the file) and so should operate on the same object. When calling 'mytable.foo(x,y)' it's written plainly that you're calling a member of a different object.

3.8.3 Binding an environment to a function

while by default a squirrel function call passes as environment object 'this', the object where the function was indexed from. However, is also possible to statically bind an environment to a closure using the built-in method `closure.bindenv(env_obj)`. The method `bindenv()` returns a new instance of a closure with the environment bound to it. When an environment object is bound to a function, every time the function is invoked, its 'this' parameter will always be the previously bound environment. This mechanism is useful to implement callbacks systems similar to C# delegates.

Note: The closure keeps a weak reference to the bound environment object, because of this if the object is deleted, the next call to the closure will result in a null environment object.

3.8.4 Lambda Expressions

```
exp := '@' '(' paramlist ')' exp
```

Lambda expressions are a syntactic sugar to quickly define a function that consists of a single expression. This feature comes handy when functional programming patterns are applied, like map/reduce or passing a compare method to `array.sort()`.

here a lambda expression:

```
local myexp = @(a,b) a + b
```

that is equivalent to:

```
local myexp = function(a,b) { return a + b; }
```

a more useful usage could be:

```
local arr = [2,3,5,8,3,5,1,2,6];
arr.sort(@(a,b) a <=> b);
arr.sort(@(a,b) -(a <=> b));
```

that could have been written as:

```
local arr = [2,3,5,8,3,5,1,2,6];
arr.sort(function(a,b) { return a <=> b; } );
arr.sort(function(a,b) { return -(a <=> b); } );
```

other than being limited to a single expression lambdas support all features of regular functions. in fact are implemented as a compile time feature.

3.8.5 Free Variables

A free variable is a variable external from the function scope as is not a local variable or parameter of the function. Free variables reference a local variable from a outer scope. In the following example the variables ‘testy’, ‘x’ and ‘y’ are bound to the function ‘foo’:

```
local x=10,y=20
local testy="I'm testy"

function foo(a,b)
{
    ::print(testy);
    return a+b+x+y;
}
```

A program can read or write a free variable.

3.8.6 Tail Recursion

Tail recursion is a method for partially transforming a recursion in a program into an iteration: it applies when the recursive calls in a function are the last executed statements in that function (just before the return). If this happens the squirrel interpreter collapses the caller stack frame before the recursive call; because of that very deep recursions are possible without risk of a stack overflow.:

```
function loopy(n)
{
    if(n>0){
        ::print("n="+n+"\n");
        return loopy(n-1);
    }
}

loopy(1000);
```

3.9 Classes

Squirrel implements a class mechanism similar to languages like Java/C++/etc... however because of its dynamic nature it differs in several aspects. Classes are first class objects like integer or strings and can be stored in table slots local variables, arrays and passed as function parameters.

3.9.1 Class Declaration

A class object is created through the keyword 'class'. The class object follows the same declaration syntax of a table(see [Tables](#)) with the only difference of using ';' as optional separator rather than ','.

For instance:

```
class Foo {
  //constructor
  constructor(a)
  {
    testy = ["stuff",1,2,3,a];
  }
  //member function
  function PrintTesty()
  {
    foreach(i,val in testy)
    {
      ::print("idx = "+i+" = "+val+" \n");
    }
  }
  //property
  testy = null;
}
```

the previous code example is a syntactic sugar for:

```
Foo <- class {
  //constructor
  constructor(a)
  {
    testy = ["stuff",1,2,3,a];
  }
  //member function
  function PrintTesty()
  {
    foreach(i,val in testy)
    {
      ::print("idx = "+i+" = "+val+" \n");
    }
  }
  //property
  testy = null;
}
```

in order to emulate namespaces, it is also possible to declare something like this:

```
//just 2 regular nested tables
FakeNamespace <- {
```

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```
    Utils = {}
}

class FakeNamespace.Utils.SuperClass {
  constructor()
  {
    ::print("FakeNamespace.Utils.SuperClass")
  }
  function DoSomething()
  {
    ::print("DoSomething()")
  }
}

function FakeNamespace::Utils::SuperClass::DoSomethingElse()
{
  ::print("FakeNamespace::Utils::SuperClass::DoSomethingElse()")
}

local testy = FakeNamespace.Utils.SuperClass();
testy.DoSomething();
testy.DoSomethingElse();
```

After its declaration, methods or properties can be added or modified by following the same rules that apply to a table(operator <-):

```
//adds a new property
Foo.stuff <- 10;

//modifies the default value of an existing property
Foo.testy <- "I'm a string";

//adds a new method
function Foo::DoSomething(a,b)
{
  return a+b;
}
```

After a class is instantiated is no longer possible to add new properties however is possible to add or replace methods.

Static variables

Squirrel's classes support static member variables. A static variable shares its value between all instances of the class. Statics are declared by prefixing the variable declaration with the keyword `static`; the declaration must be in the class body.

Note: Statics are read-only.

```
class Foo {
  constructor()
  {
    //..stuff
  }
}
```

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```

name = "normal variable";
//static variable
static classname = "The class name is foo";
};

```

Class Attributes

Classes allow to associate attributes to it's members. Attributes are a form of metadata that can be used to store application specific informations, like documentations strings, properties for IDEs, code generators etc... Class attributes are declared in the class body by preceding the member declaration and are delimited by the symbol `</` and `/>`. Here an example:

```

class Foo </ test = "I'm a class level attribute" />{
  </ test = "freakin attribute" /> //attributes of PrintTesty
  function PrintTesty()
  {
    foreach(i,val in testy)
    {
      ::print("idx = "+i+" = "+val+" \n");
    }
  }
  </ flippy = 10 , second = [1,2,3] /> //attributes of testy
  testy = null;
}

```

Attributes are, matter of fact, a table. Squirrel uses `</ />` syntax instead of curly brackets `{}` for the attribute declaration to increase readability.

This means that all rules that apply to tables apply to attributes.

Attributes can be retrieved through the built-in function `classobj.getattributes(membername)` (see [built-in functions](#)), and can be modified/added through the built-in function `classobj.setattributes(membername, val)`.

the following code iterates through the attributes of all Foo members.:

```

foreach(member, val in Foo)
{
  ::print(member+"\n");
  local attr;
  if((attr = Foo.getattributes(member)) != null) {
    foreach(i,v in attr)
    {
      ::print("\t"+i+" = "+(typeof v)+"\n");
    }
  }
  else {
    ::print("\t<no attributes>\n")
  }
}

```

3.9.2 Class Instances

The class objects inherits several of the table's feature with the difference that multiple instances of the same class can be created. A class instance is an object that share the same structure of the table that created it but holds its own values. Class *instantiation* uses function notation. A class instance is created by calling a class object. Can be useful to imagine a class like a function that returns a class instance.:

```
//creates a new instance of Foo
local inst = Foo();
```

When a class instance is created its member are initialized *with the same value* specified in the class declaration. The values are copied verbatim, *no cloning is performed* even if the value is a container or a class instances.

Note: FOR C# and Java programmers:

Squirrel doesn't clone member's default values nor executes the member declaration for each instance(as C# or java).

So consider this example:

```
class Foo {
    myarray = [1,2,3]
    mytable = {}
}

local a = Foo();
local b = Foo();
```

In the snippet above both instances will refer to the same array and same table.To achieve what a C# or Java programmer would expect, the following approach should be taken.

```
class Foo {
    myarray = null
    mytable = null
    constructor()
    {
        myarray = [1,2,3]
        mytable = {}
    }
}

local a = Foo();
local b = Foo();
```

When a class defines a method called 'constructor', the class instantiation operation will automatically invoke it for the newly created instance. The constructor method can have parameters, this will impact on the number of parameters that the *instantiation operation* will require. Constructors, as normal functions, can have variable number of parameters (using the parameter ...).:

```
class Rect {
    constructor(w,h)
    {
        width = w;
        height = h;
    }
    x = 0;
    y = 0;
    width = null;
```

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```

    height = null;
}

//Rect's constructor has 2 parameters so the class has to be 'called'
//with 2 parameters
local rc = Rect(100,100);

```

After an instance is created, its properties can be set or fetched following the same rules that apply to tables. Methods cannot be set.

Instance members cannot be removed.

The class object that created a certain instance can be retrieved through the built-in function `instance.getclass()` (see *built-in functions*)

The operator `instanceof` tests if a class instance is an instance of a certain class.:

```

local rc = Rect(100,100);
if(rc instanceof ::Rect) {
    ::print("It's a rect");
}
else {
    ::print("It isn't a rect");
}

```

Note: Since Squirrel 3.x `instanceof` doesn't throw an exception if the left expression is not a class, it simply fails

3.9.3 Inheritance

Squirrel's classes support single inheritance by adding the keyword `extends`, followed by an expression, in the class declaration. The syntax for a derived class is the following:

```

class SuperFoo extends Foo {
    function DoSomething() {
        ::print("I'm doing something");
    }
}

```

When a derived class is declared, Squirrel first copies all base's members in the new class then proceeds with evaluating the rest of the declaration.

A derived class inherit all members and properties of it's base, if the derived class overrides a base function the base implementation is shadowed. It's possible to access a overridden method of the base class by fetching the method from through the 'base' keyword.

Here an example:

```

class Foo {
    function DoSomething() {
        ::print("I'm the base");
    }
};

class SuperFoo extends Foo {
    //overridden method
    function DoSomething() {

```

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```
//calls the base method
base.DoSomething();
::print("I'm doing something");
}
}
```

Same rule apply to the constructor. The constructor is a regular function (apart from being automatically invoked on construction).:

```
class BaseClass {
    constructor()
    {
        ::print("Base constructor\n");
    }
}

class ChildClass extends BaseClass {
    constructor()
    {
        base.constructor();
        ::print("Child constructor\n");
    }
}

local test = ChildClass();
```

The base class of a derived class can be retrieved through the built-in method `getbase()`.:

```
local thebaseclass = SuperFoo.getbase();
```

Note that because methods do not have special protection policies when calling methods of the same objects, a method of a base class that calls a method of the same class can end up calling a overridden method of the derived class.

A method of a base class can be explicitly invoked by a method of a derived class though the keyword `base` (as in `base.MyMethod()`).:

```
class Foo {
    function DoSomething() {
        ::print("I'm the base");
    }
    function DoIt()
    {
        DoSomething();
    }
};

class SuperFoo extends Foo {
    //overridden method
    function DoSomething() {
        ::print("I'm the derived");
    }
    function DoIt() {
        base.DoIt();
    }
}
```

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```
//creates a new instance of SuperFoo
local inst = SuperFoo();

//prints "I'm the derived"
inst.DoIt();
```

3.9.4 Metamethods

Class instances allow the customization of certain aspects of their semantics through metamethods (see *Metamethods*). For C++ programmers: “metamethods behave roughly like overloaded operators”. The metamethods supported by classes are `_add`, `_sub`, `_mul`, `_div`, `_unm`, `_modulo`, `_set`, `_get`, `_typeof`, `_nexti`, `_cmp`, `_call`, `_delslot`, `_tostring`

Class objects instead support only 2 metamethods: `_newmember` and `_inherited`

the following example shows how to create a class that implements the metamethod `_add`:

```
class Vector3 {
  constructor(...)
  {
    if(vargv.len() >= 3) {
      x = vargv[0];
      y = vargv[1];
      z = vargv[2];
    }
  }
  function _add(other)
  {
    return ::Vector3(x+other.x,y+other.y,z+other.z);
  }

  x = 0;
  y = 0;
  z = 0;
}

local v0 = Vector3(1,2,3)
local v1 = Vector3(11,12,13)
local v2 = v0 + v1;
::print(v2.x+", "+v2.y+", "+v2.z+"\n");
```

Since version 2.1, classes support 2 metamethods `_inherited` and `_newmember`. `_inherited` is invoked when a class inherits from the one that implements `_inherited`. `_newmember` is invoked for each member that is added to the class (at declaration time).

3.10 Generators

A function that contains a `yield` statement is called '*generator function*'. When a generator function is called, it does not execute the function body, instead it returns a new suspended generator. The returned generator can be resumed through the `resume` statement while it is alive. The `yield` keyword, suspends the execution of a generator and optionally returns the result of an expression to the function that resumed the generator. The generator dies when it returns, this can happen through an explicit `return` statement or by exiting the function body; If an unhandled exception (or runtime error) occurs while a generator is running, the generator will automatically die. A dead generator cannot be resumed anymore.:

```
function geny(n)
{
    for(local i=1;i<=n;i+=1)
        yield i;
    return null;
}

local gtor=geny(10);
local x;
while(x=resume gtor) print(x+"\n");
```

the output of this program will be:

```
1
2
3
4
5
6
7
8
9
10
```

generators can also be iterated using the `foreach` statement. When a generator is evaluated by `foreach`, the generator will be resumed for each iteration until it returns. The value returned by the `return` statement will be ignored.

Note: A suspended generator will hold a strong reference to all the values stored in it's local variables except the `this` object that is only a weak reference. A running generator hold a strong reference also to the `this` object.

3.11 Constants & Enumerations

Squirrel allows to bind constant values to an identifier that will be evaluated compile-time. This is achieved through constants and Enumerations.

3.11.1 Constants

Constants bind a specific value to an identifier. Constants are similar to global values, except that they are evaluated compile time and their value cannot be changed.

constants values can only be integers, floats or string literals. No expression are allowed. are declared with the following syntax.:

```
const foobar = 100;
const floatbar = 1.2;
const stringbar = "I'm a constant string";
```

constants are always globally scoped, from the moment they are declared, any following code can reference them. Constants will shadow any global slot with the same name(the global slot will remain visible by using the :: syntax).:

```
local x = foobar * 2;
```

3.11.2 Enumerations

As Constants, Enumerations bind a specific value to a name. Enumerations are also evaluated at compile time and their value cannot be changed.

An enum declaration introduces a new enumeration into the program. Enumeration values can only be integers, floats or string literals. No expression are allowed.:

```
enum Stuff {
    first, //this will be 0
    second, //this will be 1
    third //this will be 2
}
```

or:

```
enum Stuff {
    first = 10
    second = "string"
    third = 1.2
}
```

An enum value is accessed in a manner that's similar to accessing a static class member. The name of the member must be qualified with the name of the enumeration, for example `Stuff.second` Enumerations will shadow any global slot with the same name(the global slot will remain visible by using the :: syntax).:

```
local x = Stuff.first * 2;
```

3.11.3 Implementation notes

Enumerations and Constants are a compile-time feature. Only integers, string and floats can be declared as const/enum; No expressions are allowed(because they would have to be evaluated compile time). When a const or an enum is declared, it is added compile time to the `consttable`. This table is stored in the VM shared state and is shared by the VM and all its threads. The `consttable` is a regular squirrel table; In the same way as the `roottable` it can be modified runtime. You can access the `consttable` through the built-in function `getconsttable()` and also change it through the built-in function `setconsttable()`

here some example:

```
//creates a constant
getconsttable()["something"] <- 10
//creates an enumeration
getconsttable()["somethingelse"] <- { a = "10", c = "20", d = "200"};
//deletes the constant
delete getconsttable()["something"]
//deletes the enumeration
delete getconsttable()["somethingelse"]
```

This system allows to procedurally declare constants and enumerations, it is also possible to assign any squirrel type to a constant/enumeration(function,classes etc...). However this will make serialization of a code chunk impossible.

3.12 Threads

Squirrel supports cooperative threads(also known as coroutines). A cooperative thread is a subroutine that can be suspended in mid-execution and provide a value to the caller without returning program flow, then its execution can be resumed later from the same point where it was suspended. At first look a Squirrel thread can be confused with a generator, in fact their behaviour is quite similar. However while a generator runs in the caller stack and can suspend only the local routine stack a thread has its own execution stack, global table and error handler; This allows a thread to suspend nested calls and have it's own error policies.

3.12.1 Using threads

Threads are created through the built-in function 'newthread(func)'; this function gets as parameter a squirrel function and bind it to the new thread objects (will be the thread body). The returned thread object is initially in 'idle' state. the thread can be started with the function 'threadobj.call()'; the parameters passed to 'call' are passed to the thread function.

A thread can be be suspended calling the function suspend(), when this happens the function that wokeup(or started) the thread returns (If a parameter is passed to suspend() it will be the return value of the wakeup function , if no parameter is passed the return value will be null). A suspended thread can be resumed calling the function 'threadobj.wakeup', when this happens the function that suspended the thread will return(if a parameter is passed to wakeup it will be the return value of the suspend function, if no parameter is passed the return value will be null).

A thread terminates when its main function returns or when an unhandled exception occurs during its execution.:

```
function coroutine_test(a,b)
{
    ::print(a+" "+b+"\n");
    local ret = ::suspend("suspend 1");
    ::print("the coroutine says "+ret+"\n");
    ret = ::suspend("suspend 2");
    ::print("the coroutine says "+ret+"\n");
    ret = ::suspend("suspend 3");
    ::print("the coroutine says "+ret+"\n");
    return "I'm done"
}

local coro = ::newthread(coroutine_test);

local susparam = coro.call("test","coroutine"); //starts the coroutine

local i = 1;
do
{
    ::print("suspend passed ("+susparam+")\n")
    susparam = coro.wakeup("ciao "+i);
    ++i;
}while(coro.getstatus()=="suspended")

::print("return passed ("+susparam+")\n")
```

the result of this program will be:

```
test coroutine
suspend passed (suspend 1)
the coroutine says ciao 1
suspend passed (suspend 2)
```

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```
the coroutine says ciao 2
suspend passed (suspend 3)
the coroutine says ciao 3
return passed (I'm done).
```

the following is an interesting example of how threads and tail recursion can be combined.:

```
function state1()
{
    ::suspend("state1");
    return state2(); //tail call
}

function state2()
{
    ::suspend("state2");
    return state3(); //tail call
}

function state3()
{
    ::suspend("state3");
    return state1(); //tail call
}

local statethread = ::newthread(state1)

::print(statethread.call()+"\n");

for(local i = 0; i < 10000; i++)
    ::print(statethread.wakeup()+"\n");
```

3.13 Weak References

The weak references allows the programmers to create references to objects without influencing the lifetime of the object itself. In squirrel Weak references are first-class objects created through the built-in method `obj.weakref()`. All types except null implement the `weakref()` method; however in booleans, integers, and floats the method simply returns the object itself (this because these types are always passed by value). When a weak reference is assigned to a container (table slot, array, class or instance) is treated differently than other objects; When a container slot that hold a weak reference is fetched, it always returns the value pointed by the weak reference instead of the weak reference object. This allows the programmer to ignore the fact that the value handled is weak. When the object pointed by weak reference is destroyed, the weak reference is automatically set to null.

```
local t = {}
local a = ["first", "second", "third"]
//creates a weakref to the array and assigns it to a table slot
t.thearray <- a.weakref();
```

The table slot 'thearray' contains a weak reference to an array. The following line prints "first", because tables (and all other containers) always return the object pointed by a weak ref:

```
print(t.thearray[0]);
```

the only strong reference to the array is owned by the local variable 'a', so because the following line assigns an integer to 'a' the array is destroyed:

```
a = 123;
```

When an object pointed by a weak ref is destroyed the weak ref is automatically set to null, so the following line will print "null":

```
::print(typeof(t.thearray))
```

3.13.1 Handling weak references explicitly

If a weak reference is assigned to a local variable, then it is treated as any other value:

```
local t = {}
local weakobj = t.weakref();
```

the following line prints "weakref":

```
::print(typeof(weakobj))
```

the object pointed by the weakref can be obtained through the built-in method `weakref.ref()`.

The following line prints "table":

```
::print(typeof(weakobj.ref()))
```

3.14 Delegation

Squirrel supports implicit delegation. Every table or userdata can have a parent table (delegate). A parent table is a normal table that allows the definition of special behaviors for his child. When a table (or userdata) is indexed with a key that doesn't correspond to one of its slots, the interpreter automatically delegates the get (or set) operation to its parent.:

```
Entity <- {  
}  
  
function Entity::DoStuff()  
{  
    ::print(_name);  
}  
  
local newentity = {  
    _name="I'm the new entity"  
}  
newentity.setdelegate(Entity)  
  
newentity.DoStuff(); //prints "I'm the new entity"
```

The delegate of a table can be retrieved through built-in method `table.getdelegate()`..

```
local thedelegate = newentity.getdelegate();
```

3.15 Metamethods

Metamethods are a mechanism that allows the customization of certain aspects of the language semantics. Those methods are normal functions placed in a table parent(delegate) or class declaration; It is possible to change many aspects of a table/class instance behavior by just defining a metamethod. Class objects (not instances) support only 2 metamethods `_newmember`, `_inherited`.

For example when we use relational operators other than `'=='` on 2 tables, the VM will check if the table has a method in his parent called `'_cmp'`; if so it will call it to determine the relation between the tables.:

```
local comparable={
  _cmp = function (other)
  {
    if(name<other.name)return -1;
    if(name>other.name)return 1;
    return 0;
  }
}

local a={ name="Alberto" }.setdelegate(comparable);
local b={ name="Wouter" }.setdelegate(comparable);

if(a>b)
  print("a>b")
else
  print("b<=a");
```

for classes the previous code become:

```
class Comparable {
  constructor(n)
  {
    name = n;
  }
  function _cmp(other)
  {
    if(name<other.name) return -1;
    if(name>other.name) return 1;
    return 0;
  }
  name = null;
}

local a = Comparable("Alberto");
local b = Comparable("Wouter");

if(a>b)
  print("a>b")
else
  print("b<=a");
```

3.15.1 `_set`

```
_set(idx, val)
```

invoked when the index `idx` is not present in the object or in its delegate chain. `_set` must ‘throw null’ to notify that a key wasn’t found but there were not runtime errors (clean failure). This allows the program to differentiate between a runtime error and a ‘index not found’.

3.15.2 `_get`

```
_get(idx)
```

invoked when the index `idx` is not present in the object or in its delegate chain. `_get` must ‘throw null’ to notify that a key wasn’t found but there were not runtime errors (clean failure). This allows the program to differentiate between a runtime error and a ‘index not found’.

3.15.3 `_newslot`

```
_newslot(key, value)
```

invoked when a script tries to add a new slot in a table.

if the slot already exists in the target table the method will not be invoked also if the “new slot” operator is used.

3.15.4 `_delslot`

```
_delslot(key)
```

invoked when a script deletes a slot from a table. if the method is invoked squirrel will not try to delete the slot himself

3.15.5 `_add`

```
_add(other)
```

the `+` operator

returns `this + other`

3.15.6 `_sub`

```
_sub(other)
```

the `-` operator (like `_add`)

3.15.7 `_mul`

```
_mul(other)
```

the `*` operator (like `_add`)

3.15.8 `_div`

```
_div(other)
```

the `/` operator (like `_add`)

3.15.9 `_modulo`

```
_modulo(other)
```

the `%` operator (like `_add`)

3.15.10 `_unm`

```
_unm()
```

the unary minus operator

3.15.11 `_typeof`

```
_typeof()
```

invoked by the `typeof` operator on tables, userdata, and class instances.

Returns the type of `this` as string

3.15.12 `_cmp`

```
_cmp(other)
```

invoked to emulate the `<` `>` `<=` `>=` and `<=>` operators

returns an integer as follow:

returns	relationship
<code>> 0</code>	if <code>this > other</code>
<code>0</code>	if <code>this == other</code>
<code>< 0</code>	if <code>this < other</code>

3.15.13 `_call`

```
_call(other)
```

invoked when a table, userdata, or class instance is called

3.15.14 `_cloned`

```
_cloned(original)
```

invoked when a table or class instance is cloned(in the cloned table)

3.15.15 `_nexti`

```
_nexti(previdx)
```

invoked when a userdata or class instance is iterated by a foreach loop.

If `previdx==null` it means that it is the first iteration. The function has to return the index of the 'next' value.

3.15.16 `_tostring`

```
_tostring()
```

Invoked when during string concatenation or when the `print` function prints a table, instance, or userdata. The method is also invoked by the `sq_tostring()` API.

Must return a string representation of the object.

3.15.17 `_inherited`

```
_inherited(attributes)
```

invoked when a class object inherits from the class implementing `_inherited`. The `this` contains the new class.

Return value is ignored.

3.15.18 `_newmember`

```
_newmember(index,value,attributes,isstatic)
```

invoked for each member declared in a class body (at declaration time).

If the function is implemented, members will not be added to the class.

3.16 Built-in Functions

3.16.1 Global Symbols

array(*size*[, *fill*])

creates and returns array of a specified size. If the optional parameter *fill* is specified its value will be used to fill the new array's slots. If the *fill* parameter is omitted, null is used instead.

seterrorhandler(*func*)

sets the runtime error handler

callee()

returns the currently running closure

setdebughook(*hook_func*)

sets the debug hook

enabledebuginfo(*enable*)

enable/disable the debug line information generation at compile time. *enable* != null enables. *enable* == null disables.

getroottable()

returns the root table of the VM.

setroottable(*table*)

sets the root table of the VM. And returns the previous root table.

getconsttable()

returns the const table of the VM.

setconsttable(*table*)

sets the const table of the VM; returns the previous const table.

assert(*exp*[, *message*])

throws an exception if *exp* is null or false. Throws "assertion failed" string by default, or *message* if specified.

print(*x*)

prints *x* to the standard output

error(*x*)

prints *x* in the standard error output

compilestring(*string*[, *buffername*])

compiles a string containing a squirrel script into a function and returns it:

```
local compiledscript=compilestring("::print(\"ciao\")");
//run the script
compiledscript();
```

collectgarbage()

Runs the garbage collector and returns the number of reference cycles found (and deleted). This function only works on garbage collector builds.

resurrectunreachable()

Runs the garbage collector and returns an array containing all unreachable object found. If no unreachable object is found, null is returned instead. This function is meant to help debugging reference cycles. This function only works on garbage collector builds.

type(*obj*)

return the 'raw' type of an object without invoking the metamethod '`_typeof`'.

getstackinfos(*level*)

returns the stack informations of a given call stack level. returns a table formatted as follow:

```
{
  func="DoStuff", //function name

  src="test.nut", //source file

  line=10,        //line number

  locals = {      //a table containing the local variables
    a=10,
    testy="I'm a string"
  }
}
```

level = 0 is `getstackinfos()` itself! level = 1 is the current function, level = 2 is the caller of the current function, and so on. If the stack level doesn't exist the function returns null.

newthread(*threadfunc*)

creates a new cooperative thread object(coroutine) and returns it

versionnumber

integer values describing the version of VM and compiler. e.g. for Squirrel 3.0.1 this value will be 301

version

string values describing the version of VM and compiler.

charsize

size in bytes of the internal VM representation for characters(1 for ASCII builds 2 for UNICODE builds).

intsize

size in bytes of the internal VM representation for integers(4 for 32bits builds 8 for 64bits builds).

floatsize

size in bytes of the internal VM representation for floats(4 for single precision builds 8 for double precision builds).

Default delegates

Except null and userdata every squirrel object has a default delegate containing a set of functions to manipulate and retrieve information from the object itself.

3.16.2 Integer

`integer.tofloat()`

convert the number to float and returns it

`integer.tostring()`

converts the number to string and returns it

`integer.tointeger()`

dummy function; returns the value of the integer.

`integer.tochar()`

returns a string containing a single character represented by the integer.

`integer.weakref()`

dummy function; returns the integer itself.

3.16.3 Float

`float.tofloat()`

returns the value of the float(dummy function)

`float.tointeger()`

converts the number to integer and returns it

`float.tostring()`

converts the number to string and returns it

`float.tochar()`

returns a string containing a single character represented by the integer part of the float.

`float.weakref()`

dummy function; returns the float itself.

3.16.4 Bool

`bool.tofloat()`

returns 1.0 for true 0.0 for false

`bool.tointeger()`

returns 1 for true 0 for false

`bool.tostring()`

returns "true" for true and "false" for false

bool.weakref()

dummy function; returns the bool itself.

3.16.5 String

string.len()

returns the string length

string.tointeger([*base*])

Converts the string to integer and returns it. An optional parameter base can be specified—if a base is not specified, it defaults to base 10.

string.tofloat()

converts the string to float and returns it

string.tostring()

returns the string (really, a dummy function)

string.slice(*start* [, *end*])

returns a section of the string as new string. Copies from start to the end (not included). If start is negative the index is calculated as length + start, if end is negative the index is calculated as length + end. If end is omitted end is equal to the string length.

string.find(*substr* [, *startidx*])

Searches a sub string (*substr*) starting from the index *startidx* and returns the index of its first occurrence. If *startidx* is omitted the search operation starts from the beginning of the string. The function returns null if *substr* is not found.

string.tolower()

returns a lowercase copy of the string.

string.toupper()

returns an uppercase copy of the string.

string.weakref()

returns a weak reference to the object.

3.16.6 Table

table.len()

returns the number of slots contained in a table

table.rawget(*key*)

tries to get a value from the slot 'key' without employing delegation

table.rawset(*key*, *val*)

Sets the slot 'key' with the value 'val' without employing delegation. If the slot does not exist, it will be created. Returns table itself.

table.rawdelete()

Deletes the slot key without employing delegation and returns its value. If the slot does not exist, returns null.

table.rawin(*key*)

returns true if the slot 'key' exists. the function has the same effect as the operator 'in' but does not employ delegation.

table.weakref()

returns a weak reference to the object.

table.tostring()

Tries to invoke the `_tostring` metamethod. If that fails, it returns "(table : pointer)".

table.clear()

removes all the slots from the table. Returns table itself.

table.setdelegate(*table*)

Sets the delegate of the table. To remove a delegate, 'null' must be passed to the function. The function returns the table itself (e.g. `a.setdelegate(b)` – in this case 'a' is the return value).

table.getdelegate()

returns the table's delegate or null if no delegate was set.

table.filter(*func(key, val)*)

Creates a new table with all values that pass the test implemented by the provided function. In detail, it creates a new table, invokes the specified function for each key-value pair in the original table; if the function returns 'true', then the value is added to the newly created table at the same key.

table.keys()

returns an array containing all the keys of the table slots.

table.values()

returns an array containing all the values of the table slots.

3.16.7 Array

array.len()

returns the length of the array

array.append(*val*)

appends the value 'val' at the end of the array. Returns array itself.

array.push(*val*)

appends the value 'val' at the end of the array. Returns array itself.

array.extend(*array*)

Extends the array by appending all the items in the given array. Returns array itself.

array.pop()

removes a value from the back of the array and returns it.

array.top()

returns the value of the array with the higher index

array.insert(*idx*, *val*)

inserts the value ‘val’ at the position ‘idx’ in the array. Returns array itself.

array.remove(*idx*)

removes the value at the position ‘idx’ in the array and returns its value.

array.resize(*size*[, *fill*])

Resizes the array. If the optional parameter ‘fill’ is specified, its value will be used to fill the new array’s slots when the size specified is bigger than the previous size. If the fill parameter is omitted, null is used instead. Returns array itself.

array.sort([*compare_func*])

Sorts the array in-place. A custom compare function can be optionally passed. The function prototype as to be the following.:

```
function custom_compare(a,b)
{
    if(a>b) return 1
    else if(a<b) return -1
    return 0;
}
```

a more compact version of a custom compare can be written using a lambda expression and the operator <=>

```
arr.sort(@(a,b) a <=> b);
```

Returns array itself.

array.reverse()

reverse the elements of the array in place. Returns array itself.

array.slice(*start*[, *end*])

Returns a section of the array as new array. Copies from start to the end (not included). If start is negative the index is calculated as length + start, if end is negative the index is calculated as length + end. If end is omitted end is equal to the array length.

array.weakref()

returns a weak reference to the object.

array.toString()

returns the string “(array : pointer)”.

array.clear()

removes all the items from the array

array.map(*func*(*item_value*, [*item_index*], [*array_ref*]))

Creates a new array of the same size. For each element in the original array invokes the function ‘func’ and assigns the return value of the function to the corresponding element of the newly created array. Provided func can accept up to 3 arguments: array item value (required), array item index (optional), reference to array itself (optional).

array.apply(*func*([*item_value*], [*item_index*], [*array_ref*]))

for each element in the array invokes the function ‘func’ and replace the original value of the element with the return value of the function.

array.reduce(*func*(*prevval*, *curval*)[, *initializer*])

Reduces an array to a single value. For each element in the array invokes the function ‘func’ passing the initial value (or value from the previous callback call) and the value of the current element. The return value of the function is then used as ‘prevval’ for the next element. If the optional initializer is present, it is placed before the items of the array in the calculation, and serves as a default when the sequence is empty. If initializer is not given then for sequence contains only one item, reduce() returns the first item, and for empty sequence returns null.

Given an sequence with 2 or more elements (including initializer) calls the function with the first two elements as the parameters, gets that result, then calls the function with that result and the third element, gets that result, calls the function with that result and the fourth parameter and so on until all element have been processed. Finally, returns the return value of the last invocation of func.

array.filter(*func*(*index*, *val*))

Creates a new array with all elements that pass the test implemented by the provided function. In detail, it creates a new array, for each element in the original array invokes the specified function passing the index of the element and it’s value; if the function returns ‘true’, then the value of the corresponding element is added on the newly created array.

array.find(*value*)

Performs a linear search for the value in the array. Returns the index of the value if it was found null otherwise.

3.16.8 Function

function.call(*_this*, *args...*)

calls the function with the specified environment object(‘this’) and parameters

function.pcall(*_this*, *args...*)

calls the function with the specified environment object(‘this’) and parameters, this function will not invoke the error callback in case of failure(pcall stays for ‘protected call’)

function.acall(*array_args*)

calls the function with the specified environment object(‘this’) and parameters. The function accepts an array containing the parameters that will be passed to the called function. Where *array_args* has to contain the required ‘this’ object at the [0] position.

function.pacall(*array_args*)

calls the function with the specified environment object(‘this’) and parameters. The function accepts an array containing the parameters that will be passed to the called function. Where *array_args* has to contain the required ‘this’ object at the [0] position. This function will not invoke the error callback in case of failure(pacall stays for ‘protected array call’)

function.weakref()

returns a weak reference to the object.

function.tostring()

returns the string “(closure : pointer)”.

function.setroot(*table*)

sets the root table of a closure

function.getroot()

returns the root table of the closure

function.bindenv(env)

clones the function(aka closure) and bind the environment object to it(table,class or instance). the this parameter of the newly create function will always be set to env. Note that the created function holds a weak reference to its environment object so cannot be used to control its lifetime.

function.getinfos()

returns a table containing informations about the function, like parameters, name and source name;

```
//the data is returned as a table is in form
//pure squirrel function
{
    native = false
    name = "zefuncname"
    src = "/something/something.nut"
    parameters = ["a","b","c"]
    defparams = [1,"def"]
    varargs = 2
}
//native C function
{
    native = true
    name = "zefuncname"
    paramscheck = 2
    typecheck = [83886082,83886384] //this is the typemask (see C defines OT_INTEGER,OT_
    ↪FLOAT etc...)
}
```

3.16.9 Class

class.instance()

returns a new instance of the class. this function does not invoke the instance constructor. The constructor must be explicitly called (eg. class_inst.constructor(class_inst)).

class.getattributes(membername)

returns the attributes of the specified member. if the parameter member is null the function returns the class level attributes.

class.setattributes(membername, attr)

sets the attribute of the specified member and returns the previous attribute value. if the parameter member is null the function sets the class level attributes.

class.rawin(key)

returns true if the slot 'key' exists. the function has the same effect as the operator 'in' but does not employ delegation.

class.weakref()

returns a weak reference to the object.

class.tostring()

returns the string "(class : pointer)".

class.rawget(key)

tries to get a value from the slot 'key' without employing delegation

class.rawset(key, val)

sets the slot 'key' with the value 'val' without employing delegation. If the slot does not exists, it will be created.

class.newmember(key, val[, attrs][, bstatic])

sets/adds the slot 'key' with the value 'val' and attributes 'attrs' and if present invokes the `_newmember` metamethod. If bstatic is true the slot will be added as static. If the slot does not exists , it will be created.

class.rawnewmember(key, val[, attrs][, bstatic])

sets/adds the slot 'key' with the value 'val' and attributes 'attrs'. If bstatic is true the slot will be added as static. If the slot does not exist, it will be created. It doesn't invoke any metamethod.

3.16.10 Class Instance

instance.getclass()

returns the class that created the instance.

instance.rawin(key)

returns true if the slot 'key' exists. the function has the same effect as the operator 'in' but does not employ delegation.

instance.weakref()

returns a weak reference to the object.

instance.tostring()

tries to invoke the `_tostring` metamethod, if failed. returns "(instance : pointer)".

instance.rawget(key)

tries to get a value from the slot 'key' without employing delegation

instance.rawset(key, val)

sets the slot 'key' with the value 'val' without employing delegation. If the slot does not exists, it will be created.

3.16.11 Generator

generator.getstatus()

returns the status of the generator as string : "running", "dead" or "suspended".

generator.weakref()

returns a weak reference to the object.

generator.tostring()

returns the string "(generator : pointer)".

3.16.12 Thread

thread.call(...)

starts the thread with the specified parameters

thread.wakeup(*[wakeupval]*)

wakes up a suspended thread, accepts a optional parameter that will be used as return value for the function that suspended the thread(usually `suspend()`)

thread.wakeupthrow(*objtothrow**[, propagateerror = true]*)

wakes up a suspended thread, throwing an exception in the awaken thread, throwing the object 'objtothrow'.

thread.getstatus()

returns the status of the thread ("idle","running","suspended")

thread.weakref()

returns a weak reference to the object.

thread.tostring()

returns the string "(thread : pointer)".

thread.getstackinfos(*stacklevel*)

returns the stack frame informations at the given stack level (0 is the current function 1 is the caller and so on).

3.16.13 Weak Reference

weakreference.ref()

returns the object that the weak reference is pointing at; null if the object that was point at was destroyed.

weakreference.weakref()

returns a weak reference to the object.

weakreference.tostring()

returns the string "(weakref : pointer)".

EBLINK SCRIPTING

4.1 Introduction

EBlink uses [Squirrel](#) as its scripting language. Scripts drive the device-specific behaviour: they set up the memory map, implement flash erase and write algorithms, handle custom resets, and can interact with the GDB monitor interface.

A C/C++ background is assumed throughout this chapter. Squirrel syntax is covered in the language reference; this chapter focuses on what EBLink exposes to scripts and how scripts integrate with the EBLink runtime.

4.1.1 How scripts are loaded

Scripts are specified on the command line with `-S`. Multiple `-S` options are allowed; each file is loaded in order into the same Squirrel VM so all scripts share one global namespace and see each other's definitions:

```
EBlink -I stlink -S stm32.script -G
EBlink -I stlink -S mydevice -S myReset.scr -G
```

Script files are searched in the paths added with `-P`. Multiple `-P` options accumulate search paths:

```
EBlink -I stlink -P C:\eblink\scripts -S stm32 -G
```

4.1.2 CLI defines

Key/value pairs can be passed to scripts at startup with `-D`. Multiple `-D` options are allowed:

```
EBlink -I stlink -S stm32 -D FLASH_SIZE=1024 -D RAM_SIZE=16 -G
```

Inside the script these are available as global integer constants:

```
printf("Flash speed: %d KHz\n", FLASH_SPEED);
```

4.1.3 API objects

`InterfAPI`, `DeviceAPI`, `TargetAPI`, and `ServicesAPI` are classes registered by EBLink. A script instantiates them itself, typically at the top of the main script as globals using the `<-` operator:

```
intrfApi <- InterfAPI()
targetApi <- TargetAPI()
```

These globals are then accessible to all scripts loaded via `require` or `include`. `DeviceAPI()` is usually called inline where the memory map needs to be set.

4.1.4 Minimal example

The smallest useful top-level script: instantiate the API objects, set the memory map, and define flash entry points. Flash programming requires device-specific register sequences — the body of each function is device-specific and omitted here:

```
intrfApi <- InterfAPI()
targetApi <- TargetAPI()

function ebMain()
{
    intrfApi.setSpeed(4000);
```

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```
DeviceAPI().memmap( @"<?xml version=\"1.0\"?>
<memory-map>
  <memory type=\"ram\" start=\"0x20000000\" length=\"0x8000\"/>
  <memory type=\"flash\" start=\"0x08000000\" length=\"0x40000\">
    <property name=\"pagesize\">0x400</property>
  </memory>
</memory-map>" );
}

function ebFlashStart()
{
    // Unlock flash, reset target to known state.
    return ERROR_OK;
}

function ebFlashErase(sector, address)
{
    // Device-specific sector erase via intrfApi.write(...).
    return ERROR_OK;
}

function ebFlashWrite(sector, address, buffer)
{
    // Device-specific flash write via intrfApi.writeBuf(...).
    return ERROR_OK;
}

function ebFlashDone()
{
    // Re-lock flash.
    return ERROR_OK;
}
```

4.2 Error Handling

EBlink uses Squirrel exceptions for all error signalling. Understanding the convention is essential because every API call can throw.

4.2.1 Exception types

API methods throw one of two exception types:

Integer exception — a negative `ERROR_XXX` constant. Thrown by hardware operations when the probe reports an error (target lost, flash failure, etc.):

```
try {
    local value = intrfApi.read(0x20000000);
}
catch(e) {
    if(isInteger(e))
        fprintf("Hardware error %d: %s\n", e, errorToStr(e));
}
```

String exception — a human-readable message. Thrown for usage errors: wrong argument type, buffer overflow, NULL instance, etc.:

```
try {
    local buf = EBbuffer(64);
    buf.get(200);           // offset beyond buffer
}
catch(e) {
    if(isString(e))
        fprintf("Script error: %s\n", e);
}
```

4.2.2 Return values

Functions that succeed return `ERROR_OK` (0) or a positive value (e.g. byte count). A negative return value is always an `ERROR_XXX` code. The convention is that you either throw on failure or return the code — never both:

```
function ebFlashWrite(sector, address, buffer)
{
    try {
        intrfApi.writeBuf(buffer, address);
    }
    catch(e) {
        if(isString(e)) { fprintf(e); return ERROR_NOTIFIED; }
        if(isInteger(e)) { return e; }
    }
    return ERROR_OK;
}
```

4.2.3 Propagating errors

Use `ERROR_NOTIFIED` (-1) when you have already printed an error message and want Eblink to abort without printing a second one:

```
errorf("Flash unlock failed at 0x%08X\n", addr);
return ERROR_NOTIFIED;
```

For errors the script did not handle, re-throw so Eblink can report them:

```
catch(e) {
    if(catchString(e)) return ERROR_NOTIFIED; // string: already printed
    throw e;                               // integer: let Eblink handle
}
```

4.2.4 Helper functions

catchString(value[, throwValue])

If *value* is a string exception, prints it as an error and returns `true`. If *throwValue* is supplied and *value* is a string, throws *throwValue* after printing. Returns `false` if *value* is not a string.

Typical use in a catch block:

```
catch(e) {
    if(catchString(e)) return ERROR_NOTIFIED;
    throw e;
}
```

catchThrowStr(value)

If *value* is a string, re-throws it as a string exception. Has no effect when *value* is not a string. Useful to propagate string errors up through nested try/catch blocks.

errorToStr(code)

Returns a short description string for an `ERROR_xxx` integer code. Returns an empty string for unknown codes.

Example:

```
errorf("Error %d: %s\n", e, errorToStr(e));
```

4.2.5 User-defined error codes

`ERROR_USER` (-1000) is the boundary reserved for script errors. All built-in `ERROR_xxx` codes are between -1 and -999. Scripts must define their own codes **below** `ERROR_USER` to avoid collisions:

```
const FLASH_UNLOCK_FAIL = ERROR_USER - 1; // -1001
const FLASH_TIMEOUT     = ERROR_USER - 2; // -1002
const FLASH_VERIFY_FAIL = ERROR_USER - 3; // -1003
```

These constants behave exactly like built-in error codes — they can be thrown, returned, or passed to `errorToStr()`. Because `errorToStr()` does not know about script-defined codes it returns an empty string for them; print your own message before returning or throwing:

```
function unlockFlash()
{
    if(intrfApi.read(FLASH_CR) & FLASH_LOCKED) {
```

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```
    intrfApi.write(FLASH_KEYR, KEY1);
    intrfApi.write(FLASH_KEYR, KEY2);
    if(intrfApi.read(FLASH_CR) & FLASH_LOCKED) {
        errorf("Flash unlock failed\n");
        return FLASH_UNLOCK_FAIL;
    }
}
return ERROR_OK;
}
```

Note: Never define script error codes between -1 and -999 — that range belongs to EBlint built-ins. Always subtract from ERROR_USER.

4.3 Logging & Output

EBlink routes script output through its verbosity system. Each output function maps to a verbosity level; output is only shown when the corresponding level is active. The verbosity is set with the `-v` command-line flag.

4.3.1 Output functions

All output functions accept a `printf`-style format string followed by optional arguments. They do not return a value.

printf(fmt, ...)

Default script output. Visible at `-v4` and above (`SCRIPT_INFO`).

userf(fmt, ...)

User-facing output. Visible at `-v2` and above (`USER_INFO`). For messages relevant to the end user.

noticef(fmt, ...)

Notice-level output. Visible at `-v3` and above (`NOTICE_INFO`).

warnf(fmt, ...)

Warning output. Visible at `-v6` and above (`WARN_INFO`).

errorf(fmt, ...)

Error output. Always visible regardless of verbosity level.

debugf(fmt, ...)

Debug output. Visible at `-v7` and above (`DEBUG_INFO`). For detailed internal state during script development.

tracef(fmt, ...)

Trace output. Visible at `-v8` (`TRACE_INFO`). For fine-grained step-by-step tracing.

yesNof(default, fmt, ...)

Presents an interactive yes/no question. `default` is returned when no interactive console is available: 1 for yes, 0 for no. Returns the user's choice or `default`:

```
local answer = yesNof(0, "Erase entire chip? [y/n] ");
if(answer)
    device_chip_erase();
```

4.3.2 Verbosity levels

Flag	Constant	Functions visible
<code>-v2</code>	<code>USER_INFO</code>	<code>userf</code>
<code>-v3</code>	<code>NOTICE_INFO</code>	<code>noticef</code>
<code>-v4</code>	<code>SCRIPT_INFO</code>	<code>printf</code>
<code>-v6</code>	<code>WARN_INFO</code>	<code>warnf</code>
<code>-v7</code>	<code>DEBUG_INFO</code>	<code>debugf</code>, <code>brkPt</code>
<code>-v8</code>	<code>TRACE_INFO</code>	<code>tracef</code>, <code>line step</code>

`errorf` is always shown. `-v1` shows only the EBLink header.

4.3.3 Format strings

Standard C `printf` conventions: `%d`, `%u`, `%x`, `%X`, `%f`, `%s`, `%c`. Width and precision modifiers are supported. Use `%08X` for addresses (no `%p`):

```
printf("Sector %d at 0x%08X (%d bytes)\n", sector, address, buffer.getSize());
```

4.4 Global Functions

These functions are available in all EBLink scripts without any import or declaration.

4.4.1 Script loading

include(*filename*)

Loads and executes a Squirrel script file. Searches the configured script paths. Does nothing and returns `ERROR_OK` if the file is not found — use `require` when the file must exist.

require(*filename*)

Loads and executes a Squirrel script file. Throws a string exception if the file cannot be found or loaded. Use for dependencies that must be present.

4.4.2 Type checking

isString(*value*)

Returns `true` if *value* is a string.

isInteger(*value*)

Returns `true` if *value* is an integer.

isnull(*instance*)

Returns `true` if *instance* is a null object instance. Used to check whether an API object reference is valid.

isScriptObject(*name*)

Returns `true` if a global function or object with the given name exists in the script VM. Useful before calling an optional entry point:

```
if(isScriptObject("myCustomInit"))
    myCustomInit();
```

4.4.3 Data conversion

strToNumber(*str*)

Parses a decimal or hexadecimal (`0x` prefix) string and returns the value as an integer. Returns `0` for invalid input.

Example:

```
local addr = strToNumber("0x08004000");
```

Integer2Float(*value*)

Reinterprets the bit pattern of a 32-bit integer as an IEEE 754 float. Equivalent to a C `*(float*)&value` cast. Useful when a register or memory location holds a float value.

Float2Integer(*value*)

Reinterprets the bit pattern of a float as a 32-bit integer. Equivalent to a C `*(uint32_t*)&value` cast.

4.4.4 EBLink information

cliSwitch(*switch*)

Returns the value string of a CLI option. Recognised switches are "-I" (interface), "-T" (target), "-F" (flash), "-G" (GDB). Returns an empty string if the option was not supplied:

```
local iface_opt = cliSwitch("-I");
if(iface_opt == "")
    errorf("No interface specified\n");
```

Also used in `ebServicesSetup()` to pass the original CLI service options to `ServicesAPI().addService()` — see *ServicesAPI*.

4.4.5 String utilities

astrsep(*array*, *string*, *separator*)

Splits *string* on each occurrence of *separator* and appends the resulting tokens to *array*. Returns the number of tokens added.

Example:

```
local parts = [];
astrsep(parts, "one:two:three", ":");
// parts == ["one", "two", "three"]
```

amregex(*array*, *string*, *pattern*)

Applies the regular expression *pattern* to *string*. Appends all capture groups (or the full match if there are no groups) to *array*. Returns the number of matches appended, or 0 if no match.

Example:

```
local m = [];
if(amregex(m, "REV=1.4", "REV=([0-9.]+)"))
    printf("Revision: %s\n", m[0]);
```

4.4.6 System utilities

Sleep(*ms*)

Suspends execution for *ms* milliseconds.

GetTickCount()

Returns the number of milliseconds elapsed since EBLink started. Useful for implementing timeouts:

```
local t0 = GetTickCount();
while(GetTickCount() - t0 < 500) {
    if(target.getState() == TARGET_HALTED) break;
    Sleep(10);
}
```

AnimateCursor()

Starts a spinning cursor animation on the console. Call before a long operation.

AnimateDone()

Stops the cursor animation started by `AnimateCursor()`.

4.4.7 Math

Each function takes two values of the appropriate type and returns the smaller or larger of the two.

FL_MIN(*a*, *b*)

FL_MAX(*a*, *b*)

Float minimum / maximum.

UN_MIN(*a*, *b*)

UN_MAX(*a*, *b*)

Unsigned integer minimum / maximum.

IN_MIN(*a*, *b*)

IN_MAX(*a*, *b*)

Integer minimum / maximum.

4.5 Constants

All constants listed here are available in every EBLink script without declaration.

4.5.1 VERSION

VERSION is a float containing the EBLink API version:

```
if(VERSION < 3.0)
    errorf("This script requires EBLink API 3.0 or newer\n");
```

4.5.2 Target state

Returned by `TargetAPI.getState()`.

Constant	Value	Meaning
TARGET_UNKNOWN	0	State cannot be determined
TARGET_RUNNING	1	CPU is executing code
TARGET_HALTED	2	CPU is halted (debug state)
TARGET_RESET	3	CPU is held in reset

4.5.3 Reset types

Passed to `TargetAPI.reset()`.

Constant	Value	Effect
SYSTEM_RESET	0	Full system reset via AIRCR.SYSRESETREQ
CORE_RESET	1	Core-only reset via AIRCR.VECTRESET
JTAG_RESET	2	Hardware reset via the nRESET pin
USER_RESET	3	Custom reset — calls <code>ebResetUser()</code>

4.5.4 Error codes

All negative integers thrown by API methods correspond to one of these constants. Scripts can compare a caught integer exception against them:

```
catch(e) {
    if(e == ERROR_TARGET_LOST) { errorf("Target disconnected\n"); return e; }
    throw e;
}
```

User-defined error codes must be more negative than `ERROR_USER`:

```
const MY_UNLOCK_FAIL = (ERROR_USER - 1); // -1001
```

Constant	Value	Description
ERROR_OK	0	No error
ERROR_NOTIFIED	-1	Error already reported, abort silently
ERROR_SYNTAX	-2	Script syntax error
ERROR_OUT_MEMM	-3	Out of memory
ERROR_TARGET_LOST	-4	Target connection lost
ERROR_SOCKET_CLOSE	-5	GDB socket closed
ERROR_SOCKET_ERR	-6	GDB socket error
ERROR_NO_DEVICE	-7	No device found
ERROR_OUT_OF_RANGE	-8	Value out of range
INTERFACE_NOT_SUPPORTED	-20	Feature not supported by this interface
INTERFACE_WAIT	-21	Interface speed too high
INTERFACE_ERROR	-22	Interface communication error
ERROR_INTERNAL	-23	Internal EBlind error
ERROR_NOT_FOUND	-100	Requested item not found
ERROR_UNLOCK	-300	Flash unlock failed
ERROR_VOLTAGE	-301	Target voltage out of range
ERROR_FLASH	-302	Flash operation failed
ERROR_TIMEOUT	-303	Operation timed out
ERROR_USER	-1000	Base for user-defined error codes

INTERFACE_WAIT (-21) usually means the probe speed set with `iface.setSpeed()` is too high for the target. Try reducing it or use the `-D FLASH_SPEED=<KHz>` CLI option.

4.6 EBLink Interaction

EBlink calls specific functions in the script at defined points in its operation. Functions not defined in the script are silently skipped, except where marked mandatory.

4.6.1 Overview

Function	Required	Called when
ebMain()	mandatory	All scripts loaded, CLI defines set
ebFlashStart()	optional	Flash operation is about to begin
ebFlashErase(sector_nr, sector_addr)	optional	A sector must be erased before writing
ebFlashWrite(sector_nr, addr, buf)	mandatory	A sector must be written
ebFlashDone()	optional	All flash sectors processed
ebFlashEraseChip()	optional	Full chip erase requested (-F erase)
ebResetUser()	optional*	A USER_RESET is requested
ebResetPost(reset_type)	optional	After every reset, regardless of type
ebServicesSetup()	optional	EBlink service initialisation
ebGDBqRcmd(cmd, length, buffer)	optional	GDB monitor command received

* Mandatory when the script uses USER_RESET as its reset strategy.

4.6.2 Call order

```

Script loaded
|
| v
ebMain()                <- mandatory: set memmap, initialise device
|
| +--- flash operation requested
| |
| | v
| | ebFlashStart()        <- optional: prepare hardware for flash
| |
| | +-----+
| | | for each modified sector:
| | | ebFlashErase(nr, addr) <- optional
| | | ebFlashWrite(nr, addr, buf) <- mandatory
| | |
| | +-----+
| |
| | ebFlashDone()         <- optional: clean up after flash
| |
| | +--- chip erase requested
| | |
| | | ebFlashEraseChip()  <- optional (falls back to sector loop)
| | |
| | +--- reset requested
| | |
| | | ebResetUser()        <- mandatory when USER_RESET type used
| | | ebResetPost(type)    <- optional: called after every reset

```

ebServicesSetup() is called once during service initialisation, independent of the flash or reset flow. ebGDBqRcmd() is called on every incoming GDB monitor command, independent of the other flows.

4.6.3 Reference

ebMain()

mandatory

- Called once after all scripts are loaded and all -D defines are set.
- Must call DeviceAPI().memmap() — EBLink aborts if no memory map is set.
- Typical use: detect device, set interface speed, configure flash parameters.

Returns: nothing required.

```
intrfApi <- InterfAPI()
targetApi <- TargetAPI()

function ebMain()
{
    intrfApi.setSpeed(4000);

    DeviceAPI().memmap( @@"<?xml version=\"1.0\"?>
<memory-map>
  <memory type=\"flash\" start=\"0x08000000\" length=\"0x80000\">
    <property name=\"pagesize\">0x800</property>
  </memory>
  <memory type=\"ram\" start=\"0x20000000\" length=\"0x20000\"/>
</memory-map>" );
}
```

ebFlashStart()

optional

- Called once before the first sector is touched.
- Typical use: unlock flash, reset target to known state, set interface speed.

Returns: ERROR_OK or a negative error code.

ebFlashErase(sector_nr, sector_addr)

optional

- sector_nr — zero-based sector index from the memory map.
- sector_addr — start address of the sector.
- If not defined, EBLink performs no explicit sector erase.

Returns: ERROR_OK or a negative error code.

ebFlashWrite(sector_nr, sector_addr, buffer)

mandatory when flashing

- sector_nr — zero-based sector index from the memory map.
- sector_addr — start address of the sector in target memory.
- buffer — *EBbuffer* containing the sector data, trimmed to the last non-erased byte.
- Called for every sector that needs to be written, after erasure.

Returns: ERROR_OK or a negative error code.

```
function ebFlashWrite(sector, address, buffer)
{
    try {
        intrfApi.writeBuf(buffer, address);
```

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```

    }
    catch(e) {
        if(catchString(e)) return ERROR_NOTIFIED;
        return e;
    }
    return ERROR_OK;
}

```

ebFlashDone()*optional*

- Called once after all sectors have been processed, success or failure.
- Typical use: re-lock flash, reset the device.

Returns: ERROR_OK or a negative error code.

ebFlashEraseChip()*optional*

- Called when a full chip erase is requested (-F erase option).
- Runs standalone — ebFlashStart and ebFlashDone are **not** called around it.
- If not defined, EBLink falls back to calling ebFlashStart, ebFlashErase for every sector, and ebFlashDone.

Returns: ERROR_OK or a negative error code.

ebResetUser()*mandatory when USER_RESET type is used*

- Implement a custom reset sequence — for example via a peripheral register when the nRESET pin is not available.

Returns: nothing required.

```

function ebResetUser()
{
    intrfApi.write(0xE000ED0C, 0x05FA0004); // SYSRESETREQ via AIRCR
    Sleep(50);
}

```

ebResetPost(reset_type)*optional*

- reset_type — one of the xxx_RESET constants (see [Constants](#)).
- Called after every reset, regardless of type.
- Typical use: set stack pointer and PC from the vector table, wait for the device to come out of reset.

Returns: ERROR_OK or a negative error code.

```

function ebResetPost(type)
{
    if(type != USER_RESET) {
        targetApi.writeReg("SP", intrfApi.read(0x08000000));
        targetApi.writeReg("PC", intrfApi.read(0x08000004));
    }
    return ERROR_OK;
}

```

ebServicesSetup()*optional*

- Called during EBLink service initialisation.
- If not defined, EBLink starts services as specified on the CLI.
- As soon as one `ServicesAPI().addService()` call is made, EBLink no longer starts any services automatically — the script is responsible for starting all services.
- Use `cliSwitch("-G")` to retrieve the CLI GDB option and pass it to `addService()` if needed.

Returns: `ERROR_OK` or a negative error code.

ebGDBqRcmd(cmd, length, buffer)*optional*

- Called when a GDB monitor command arrives, before EBLink's own built-in monitor handling.
- `cmd` — the monitor command string.
- `length` — length of the command string in bytes.
- `buffer` — an *EBbuffer* the script fills with the response text to send back to GDB.
- If the script handles the command: fill `buffer` and return `ERROR_OK`. EBLink sends the buffer contents back to GDB and skips built-in handling.
- If the script does not handle the command: return a negative error code or leave `buffer` empty. EBLink then falls through to its own monitor command dispatcher.

Returns: `ERROR_OK` (command handled) or a negative error code (pass through).

The two examples below show the same `ReadMemAP` command implemented with two different parsing strategies.

Parsing with `amregex` — a single regex captures all arguments at once:

```
function ebGDBqRcmd(packet, length, ebBuffer)
{
    try {
        local matches = []
        // ReadMemAP <APBaseAddr> <Addr> <NumItems> <Flags>
        if(amregex(matches, packet,
            @"ReadMemAP\s+(0x[0-9a-fA-F]+|[0-9]+)\s+(0x[0-9a-fA-F]+|[0-9]+)" +
            @"\s+(0x[0-9a-fA-F]+|[0-9]+)\s+(0x[0-9a-fA-F]+|[0-9]+)" > 0)
        {
            local APbase = strToNumber(matches[1])
            local addr    = strToNumber(matches[2])
            local num     = strToNumber(matches[3])
            local flags   = strToNumber(matches[4])

            local byteLen = (flags == 0 ? 4 : flags)
            local width   = (flags == 0 ? -1 : byteLen * 8)
            local fspec    = format("0x%%0%dX", byteLen * 2)

            local eb = EBbuffer()
            InterfAPI().readBuf(eb, addr, num * byteLen, width)
            if(width > 0) eb.access(width)

            local outStr = "O.K.:"
            for(local i = 0; i < num; i++) {
                outStr += format(fspec, eb[i])
                if(i < num - 1) outStr += ","
            }
        }
    }
}
```

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```

        ebBuffer.format(outStr + "\n")
    }
    return ERROR_OK
}
catch(e) {
    if(catchString(e)) return ERROR_NOTIFIED
    return ERROR_NOTIFIED
}
}

```

Parsing with astrsep — tokenises on spaces, then checks token by name:

```

function ebGDBqRcmd(packet, length, ebBuffer)
{
    try {
        local tokens = []
        astrsep(tokens, packet, " ")

        // ReadMemAP <APBaseAddr> <Addr> <NumItems> <Flags>
        if(tokens[0] == "ReadMemAP") {
            if(tokens.len() < 5) return ERROR_OK

            local APbase = strToNumber(tokens[1])
            local addr = strToNumber(tokens[2])
            local num = strToNumber(tokens[3])
            local flags = strToNumber(tokens[4])

            local byteLen = (flags == 0 ? 4 : flags)
            local width = (flags == 0 ? -1 : byteLen * 8)
            local fspec = format("0x%%0%dX", byteLen * 2)

            local eb = EBbuffer()
            InterfAPI().readBuf(eb, addr, num * byteLen, width)
            if(width > 0) eb.access(width)

            local outStr = "O.K.:"
            for(local i = 0; i < num; i++) {
                outStr += format(fspec, eb[i])
                if(i < num - 1) outStr += ","
            }
            ebBuffer.format(outStr + "\n")
        }
        return ERROR_OK
    }
    catch(e) {
        if(catchString(e)) return ERROR_NOTIFIED
        return ERROR_NOTIFIED
    }
}

```

Use `amregex` when the command syntax is complex or needs strict validation. Use `astrsep` when the command is simple whitespace-delimited tokens.

4.7 EBbuffer

EBbuffer is a general-purpose binary data buffer. Unlike the other API objects it can be created freely by scripts. It is the standard way to pass blocks of data to and from hardware interface functions.

An EBbuffer has a fixed size once allocated, a configurable access width (8, 16, or 32 bits) that governs how the `[]` index operator works, and an optional trim value for alignment padding.

4.7.1 Construction

EBbuffer(*[size_or_other]*)

Creates an EBbuffer. Three forms:

- **EBbuffer()** — empty, unallocated. Use **allocate()** before access, or pass directly to **readBuf()** which allocates internally.
- **EBbuffer(size)** — allocates *size* bytes, zero-initialised.
- **EBbuffer(other)** — copies another EBbuffer. Access width resets to 32 bits.

4.7.2 Memory access

EBbuffer.get(*offset* [, *width*])

Returns the value at byte *offset*. *width* is 8, 16, or 32 bits; defaults to 32. Throws a string exception if the access would exceed the buffer.

EBbuffer.set(*offset*, *value* [, *width*])

Writes *value* at byte *offset*. *width* is 8, 16, or 32 bits; defaults to 32. Throws a string exception if the access would exceed the buffer.

4.7.3 Index operator

buf[index] reads or writes one element using the current access width set by **access()**. The byte offset is *index* * (**accessWidth** / 8):

```
local buf = EBbuffer(64);
buf.access(8);           // set 8-bit access
buf[0] = 0xDE;
buf[1] = 0xAD;
printf("0x%02X 0x%02X\n", buf[0], buf[1]);
```

4.7.4 Buffer management

EBbuffer.allocate(*size*)

Allocates *size* bytes for an unallocated buffer. Throws if the buffer is already allocated or is a reference buffer. The buffer is zero-initialised.

EBbuffer.getSize()

Returns the buffer size in bytes.

EBbuffer.access(*[width]*)

Sets the access width used by the `[]` operator to *width* (8, 16, or 32). Passing `0` queries the current width without changing it. Returns the previous width.

Default access width is 32 bits.

EBbuffer.format(*fmt*, ...)

Fills the buffer with the result of a `printf`-style format operation, replacing any existing content. The buffer is re-allocated to fit the formatted string exactly. Returns the number of bytes written.

Useful for building XML or text payloads to write to target memory:

```
local buf = EBbuffer();
buf.format("BOOT=%d\n", boot_mode);
intrfApi.writeBuf(buf, 0x20001000);
```

4.7.5 Trimming

Trim is used with reference buffers (obtained from `readBuf`) to pad the logical size to an alignment boundary without allocating extra memory.

EBbuffer.trim(*boundary*)

Sets the trim so that `trimmedSize()` is rounded up to a multiple of *boundary* bytes. Only valid for reference buffers (those filled by `readBuf`). Pass `0` to clear the trim.

EBbuffer.getTrim()

Returns the current trim padding in bytes.

4.7.6 Operators

buf1 + buf2

Concatenates two buffers and returns a new **EBbuffer** containing the data of `buf1` followed by `buf2`.

buf1 <=> buf2

Compares the contents of two buffers byte by byte. Returns `0` if equal, `-1` if `buf1` is less than `buf2`, `1` if greater. Used by the Squirrel `==` and comparison operators.

4.7.7 Example

Reading a flash sector and comparing it to a file:

```
local sector = EBbuffer();
intrfApi.readBuf(sector, 0x08000000, 2048);

local expected = EBbuffer();
// ... fill expected from file ...

if((sector <=> expected) == 0)
    printf("Sector matches\n");
else
    printf("Sector differs\n");
```

4.8 InterfAPI

InterfAPI is a class registered by EBLink that represents the physical debug probe (ST-Link, CMSIS-DAP, etc.). Scripts instantiate it once at global scope and share the instance across all included files:

```
intrfApi <- InterfAPI()
```

All methods that communicate with hardware throw an integer `ERROR_XXX` exception on failure.

4.8.1 Probe information

InterfAPI.getName()

Returns the interface name string (e.g. "stlink").

InterfAPI.getTapId()

Returns the JTAG TAP ID / SWD target ID as a 32-bit integer. Returns 0 if the read fails.

InterfAPI.getSpeed()

Returns the current probe communication speed in KHz.

InterfAPI.setSpeed(speed)

Sets the probe speed to `speed` KHz. Throws on error:

```
intrfApi.setSpeed(4000);
```

InterfAPI.targetVoltage()

Returns the measured target VCC voltage as a float (volts).

InterfAPI.resetCtrl(state)

Asserts (`state = true`) or deasserts (`state = false`) the hardware nRESET line. Throws on error.

4.8.2 DAP access

These functions provide direct access to the ARM Debug Access Port (DAP) registers, bypassing the normal memory interface. Useful for reading the IDCODE register or configuring AP/DP registers during device identification.

InterfAPI.setDapAp(ap)

Selects the active Access Port (0–127). Returns the previously selected AP number. Throws on error.

InterfAPI.getDapAp()

Returns the currently selected Access Port number.

InterfAPI.readDap(ap, addr)

Reads a 32-bit value from DAP register `addr` on access port `ap`. Throws on error.

InterfAPI.writeDap(ap, addr, value)

Writes a 32-bit `value` to DAP register `addr` on access port `ap`. Throws on error.

4.8.3 Memory read

InterfAPI.read(*addr*[, *width*])

Reads a single value from target address *addr*. *width* is 8, 16, or 32 bits. When omitted (or -1), uses the probe's native 32-bit debug access path. Returns the value as an integer. Throws on error:

```
local cpuid = intrfApi.read(0xE000ED00);
printf("CPUID: 0x%08X\n", cpuid);
```

InterfAPI.readBuf(*buffer*, *addr*, *length*[, *width*])

Reads *length* bytes from target address *addr* into *buffer* (an `EBbuffer()`). The buffer is (re-)allocated to *length* bytes internally — no prior allocation is needed. *width* controls bus transfer width (8, 16, or 32); omit or pass -1 for the default. Returns *length* on success. Throws on error:

```
local buf = EBbuffer();
intrfApi.readBuf(buf, 0x08000000, 1024);
```

4.8.4 Memory write

InterfAPI.write(*addr*, *value*[, *width*])

Writes a single value to target address *addr*. *width* is 8, 16, or 32 bits; defaults to 32-bit debug access when omitted. Throws on error.

InterfAPI.writeBuf(*buffer*, *addr*[, *width*[, *offset*, *length*]])

Writes the contents of *buffer* to target address *addr*.

Optional parameters:

- *width* — bus transfer width: 8, 16, or 32. Omit or -1 for default.
- *offset* — byte offset into *buffer* to start from (must be paired with *length*).
- *length* — number of bytes to write (required when *offset* is given).

Throws on error:

```
// Write entire buffer
intrfApi.writeBuf(buf, 0x08000000);

// Write 256 bytes starting at offset 512
intrfApi.writeBuf(buf, 0x08001000, -1, 512, 256);
```

4.8.5 File loading

InterfAPI.loadBin(*filename*, *addr*)

Reads a binary file from disk and writes it to target memory starting at *addr*. EBlink searches the configured script paths for the file. Returns the number of bytes written. Throws an integer exception (via `ERROR_NOTIFIED`) if the file cannot be opened.

InterfAPI.loadString(*addr*, *str*, *len*)

Writes the raw bytes of *str* (up to *len* bytes) to target address *addr*. Returns *len*. Throws on hardware error. Useful for writing null-terminated strings or packed data to target RAM:

```
intrfApi.loadString(0x20000000, "BOOT", 4);
```

4.9 DeviceAPI

DeviceAPI is a class registered by EBLink that manages the device memory map and the flash sector cache. A new instance is created where needed:

```
DeviceAPI().memmap( ... )
```

The memory map must be set in `ebMain()`; without it EBLink cannot operate.

4.9.1 Memory map

DeviceAPI.**memmap**(*xml*)

Sets the device memory map from an XML string. The XML format is the GDB memory-map DTD. Must be called in `ebMain()`. Returns `ERROR_OK` on success or a negative error code if the XML could not be parsed.

The `pagesize` property inside a flash region defines the sector size used for the flash cache:

```
DeviceAPI().memmap( @@"<?xml version=\"1.0\"?>
    <!DOCTYPE memory-map PUBLIC \"+//IDN gnu.org//DTD GDB Memory Map V1.0//EN\"
        \"http://sourceware.org/gdb/gdb-memory-map.dtd\">
    <memory-map>
        <memory type=\"flash\" start=\"0x08000000\" length=\"0x100000\">
            <property name=\"pagesize\">0x800</property>
        </memory>
        <memory type=\"ram\" start=\"0x20000000\" length=\"0x20000\"/>
    </memory-map>" );
```

Multiple flash and RAM regions can be defined. GDB uses this information for breakpoints, stack unwinding, and the memory display.

DeviceAPI.**memmapFromFile**(*filename*)

Loads the memory map XML from a file instead of an inline string. Returns `ERROR_OK` on success or `-1` on failure (file not found, read error, or XML parse error).

4.9.2 Flash management

DeviceAPI.**isChipErase**()

Returns true if a full chip erase was requested (e.g. EBLink `-F` erase option). Use in `ebFlashStart()` to select the erase strategy:

```
function ebFlashStart()
{
    if(DeviceAPI().isChipErase())
        flash_mass_erase();
}
```

DeviceAPI.**sectorCount**()

Returns the total number of flash sectors defined in the memory map.

DeviceAPI.**setFlashEraseValue**(*value*)

Sets the byte value that EBLink considers an “erased” cell. Default is `0xFF` (standard NOR flash). Change this for devices with a different erase value (e.g. `0x00`).

DeviceAPI.setFlashDontTrim(*value*)

When *value* is `true`, disables the automatic trimming of the flash write buffer to the erased value. By default EBlink strips trailing erased bytes from the end of each sector before calling `ebFlashWrite()` to avoid unnecessary writes.

DeviceAPI.modifyFlash(*addr*, *buffer*)

Writes *buffer* directly into the EBlink flash sector cache at address *addr*, without touching the target hardware. EBlink will include these modified sectors in the next flash programming cycle. Returns `ERROR_OK` or a negative error code.

Use this to patch data in flash before the normal write cycle:

```
local patch = EBbuffer(4);
patch.set(0, 0xDEADBEEF);
DeviceAPI().modifyFlash(0x080FFFC, patch);
```

4.10 TargetAPI

TargetAPI is a class registered by EBLink that controls the CPU state: halting, resuming, reading and writing registers, executing code, and resetting the device. Scripts instantiate it once at global scope:

```
targetApi <- TargetAPI()
```

All methods throw an integer `ERROR_XXX` exception on failure.

4.10.1 Target information

TargetAPI.getName()

Returns the device name string as reported by the target driver (e.g. "STM32F4") or null if no target is attached.

TargetAPI.getCpuName()

Returns the ARM CPU core name (e.g. "M0", "M3", "M4").

TargetAPI.getState()

Returns the current CPU state as one of the `TARGET_XXX` constants: `TARGET_UNKNOWN`, `TARGET_RUNNING`, `TARGET_HALTED`, `TARGET_RESET`.

4.10.2 Execution control

TargetAPI.halt()

Requests the CPU to halt. Throws on error.

TargetAPI.resume()

Resumes a halted CPU. Throws on error.

TargetAPI.poll()

Polls the target hardware and updates the internal state. Call this after `resume()` before reading `getState()` if you need an immediate state update.

TargetAPI.execute(addr, wait)

Writes `addr` to the program counter (PC register) and resumes the CPU. If `wait` is `true`, blocks until the CPU halts again. Useful for running a small RAM-resident flash algorithm:

```
// Upload algorithm to RAM, then execute it
intrfApi.writeBuf(algo_buf, 0x20000000);
targetApi.execute(0x20000000, true);
```

TargetAPI.reset(type)

Resets the device using one of the `xxx_RESET` constants:

```
targetApi.reset(SYSTEM_RESET);
```

For `USER_RESET`, the `ebResetUser()` script function is called instead of a hardware reset. After the reset the `ebResetPost()` function is called if it exists.

4.10.3 Register access

Register names are core-specific. For ARM Cortex-M the standard names are "r0" through "r15", "pc" (same as "r15"), "sp" (same as "r13"), "lr" (same as "r14"), and "xpsr".

`TargetAPI.readReg(name)`

Returns the 32-bit value of the named register. Throws on error or if the register name is unknown:

```
local pc = targetApi.readReg("pc");
printf("PC = 0x%08X\n", pc);
```

`TargetAPI.writeReg(name, value)`

Writes value to the named register. Throws on error or if the register name is unknown:

```
targetApi.writeReg("r0", 0x00000000); // clear R0
```

4.10.4 Example: read and print all core registers

```
function dump_regs()
{
    local regs = ["r0","r1","r2","r3","r4","r5","r6","r7",
                  "r8","r9","r10","r11","r12","sp","lr","pc","xpsr"];
    foreach(name in regs)
        printf("%-6s = 0x%08X\n", name, targetApi.readReg(name));
}
```

4.11 ServicesAPI

ServicesAPI is a class registered by EBLink, used to register EBLink services from within the ebServicesSetup() script function.

4.11.1 Service start behaviour

When ebServicesSetup() is **not defined** in the script, EBLink starts services as specified on the command line (e.g. -G for a GDB server).

As soon as the script calls ServicesAPI().addService() **at least once**, EBLink no longer starts any services on its own. From that point the script is fully responsible for starting all required services.

The script can retrieve the original CLI option string with cliSwitch() and use it as the base for the service options. This is useful when the script needs to add extra parameters (such as a DAP access port) while still honouring the options the user passed on the command line.

Note: Place any script-added options **after** the cliSwitch() string so they override the CLI defaults, not the other way around.

4.11.2 Example: dual-core GDB setup

The following pattern is taken from the STM32H7 dual-core script. When two cores are present, two separate GDB servers are started — one per core — each using the original -G CLI options with an overridden DAP access port:

```
function ebServicesSetup()
{
    // Single-core: let EBLink handle services via CLI as usual
    if(!hasDualCore)
        return ERROR_OK;

    try {
        local servicesApi = :: ServicesAPI()

        //
        //          type          name          options
        servicesApi.addService("GDB-server", "GDB_CM7", format("%s,ap=%d", cliSwitch(
→ "-G"), CM7_AP))
        servicesApi.addService("GDB-server", "GDB_CM4", format("%s,ap=%d", cliSwitch(
→ "-G"), CM4_AP))
    }
    catch(e) {
        if(catchString(e)) return ERROR_NOTIFIED;
        return e;
    }

    return ERROR_OK;
}
```

The cliSwitch("-G") call returns the full -G option string as supplied on the command line (e.g. "port=4242"). The ap= parameter is appended after it so it takes precedence over any ap= value in the original CLI string.

4.11.3 Reference

`ServicesAPI.addService(type, name, options)`

Registers a service of the given `type` with the given `name` and `options` string. Throws an integer exception on failure.

- `type` — service type string (e.g. "GDB-server")
- `name` — instance name for this service
- `options` — option string passed to the service (format depends on type)

Calling this function at least once causes EBLink to hand full service control to the script — no CLI-specified services will be started automatically.

4.12 Script Debugger

EBlink includes a built-in Squirrel script debugger. It is activated through the verbosity flag and lets you trace, step through, and inspect script execution from the console without an external IDE.

4.12.1 Enabling the debugger

Flag	Effect
-v7	Enables the debug hook. EBlink prints each function call and return. <code>brkPt()</code> calls become active. Interactive stepping on call/return/exception events.
-v8	Additionally enables line-level debug info in the compiled bytecode. Stepping to the next source line becomes available. Produces more detailed output.

With -v7 the output shows the function name and event type for each call/return. With -v8 each source line is also reported.

4.12.2 Trace output format

While the debugger is active, EBlink prints a trace line for each event:

```
Squirrel debug      ebMain(c)[ 0]      {stm32.script}
Squirrel debug      ebFlashWrite(c)[ 0]    {stm32.script}
Squirrel debug      ebFlashWrite(r)[145]   {stm32.script}
```

Columns: debug/trace mode, function name (padded), event type in parentheses, line number in brackets, source file in braces.

Event types:

c	Function call (entering a function)
r	Function return (leaving a function)
e	Exception (a <code>throw</code> is about to fire)
l	Source line (only with -v8)

4.12.3 Script breakpoints

`brkPt()`

Inserts a breakpoint at the call site. Only active with -v7 or higher; has no effect otherwise. When hit, EBlink prints the call stack with local variables and pauses for console input:

```
function ebFlashErase(sector, address)
{
    brkPt();           // execution stops here under -v7
    flash_unlock();
    // ...
}
```

4.12.4 Interactive commands

At a breakpoint or step-stop, EBLink shows the call stack and prompts:

```
== BREAKPOINT ==
*FUNCTION [ebFlashErase()] stm32.script line [42]
*FUNCTION [ebMain()] stm32.script line [15]
-LOCALS:
  [sector] INTEGER( 0 )
  [address] INTEGER( 134217728 )

Go, next Call, Return or Exception? [g,c,r,e]
```

Available commands:

Key	Level	Action
g	-v7	Go: continue until the next <code>brkPt()</code> or end
c	-v7	Call: run until the next function call
r	-v7	Return: run until the current function returns
e	-v7	Exception: run until the next <code>throw</code>
l	-v8 only	Line: run until the next source line

4.12.5 Call stack and locals

At every stop, EBLink prints the full call stack from the current function up to the top level, followed by all visible local variables with their types and values. Supported types: `INTEGER`, `FLOAT`, `STRING`, `NULL`, `TABLE`, `ARRAY`, `CLOSURE`, `INSTANCE`, `USERPOINTER`.

This information is the primary tool for understanding unexpected script behaviour — check the locals before stepping further to confirm the values are what you expect.

4.12.6 Tips

- Place `brkPt()` just before the operation you suspect is wrong, not at the top of a function, to reduce the number of steps needed.
- Use `-v7` for function-level tracing during initial development; switch to `-v8 only` when you need to trace individual lines.
- The debugger pauses the entire EBLink process — a GDB session connected to the same instance will time out if you leave a breakpoint open too long.

4.13 Addendum: Squirrel Extensions

EBlink modifies the Squirrel compiler and lexer in two places to make common MCU scripting patterns more concise. These extensions are not part of the upstream Squirrel language.

4.13.1 Multiline verbatim strings

Standard Squirrel verbatim strings (`@"..."`) suppress escape processing but do not allow embedded newlines. EBlink adds a double-prefix form `@@"..."` that allows newlines inside the string literal:

```
const mem_map = @@"<?xml version=\"1.0\"?>
<memory-map>
  <memory type=\"flash\" start=\"0x08000000\" length=\"0x80000\">
    <property name=\"pagesize\">0x800</property>
  </memory>
  <memory type=\"ram\" start=\"0x20000000\" length=\"0x20000\"/>
</memory-map>";
```

The string is copied verbatim including all embedded newline characters. Backslash escapes inside `@@"..."` are not interpreted (same as `@"..."` for single-line strings). This makes it practical to embed XML memory maps and other multi-line text directly in a script.

Note: `@"..."` — single-line verbatim: no escape processing, no newlines.

`@@"..."` — multiline verbatim: no escape processing, physical newlines in source are ignored (use `\n` for a newline in the value).

4.13.2 Compile-time constant expressions

Standard Squirrel `const` and `enum` values may only be integer, float, or string literals — no expressions are permitted. EBlink extends this so that an integer expression wrapped in parentheses is evaluated at compile time:

```
const BASE    = 0x40020000;
const CR1     = (BASE + 0x00);
const CR2     = (BASE + 0x04);
const ENABLE  = (1 << 0);
const MODE    = (3 << 2);

enum Flags {
  FLAG_A = (1 << 0),
  FLAG_B = (1 << 4)
}
```

Supported operators inside the parentheses:

Operator	Operation
<<	Left shift
>>	Right shift
*	Multiply
/	Divide
+	Add
-	Subtract / unary minus

Constraints:

- Only integer operands — no floats, no strings.
- An operand may be an integer literal or the name of a previously declared integer constant.
- Exactly one operator per expression. (A + B) is valid; (A + B + C) is a compile error. Chain constants to work around this:

```
const AB  = (A + B);  
const ABC = (AB + C);
```

- Bitwise operators (|, &, ^) are **not** supported inside ().

Pseudo BNF:

```
ConstExpr : '(' Operand ')'  
          | '(' Operand Operator Operand ')'  
Operand   : ['-'] ( IntegerLiteral | ConstantName )  
Operator  : '<<' | '>>' | '*' | '/' | '+' | '-'
```

4.14 API Quick Reference

DeviceAPI

- `isChipErase()`
- `memmap()`
- `memmapFromFile()`
- `modifyFlash()`
- `sectorCount()`
- `setFlashDontTrim()`
- `setFlashEraseValue()`

EBbuffer

- `EBbuffer()` (constructor)
- `access()`
- `allocate()`
- `format()`
- `get()`
- `getSize()`
- `getTrim()`
- `set()`
- `trim()`

InterfAPI

- `getDapAp()`
- `getName()`
- `getSpeed()`
- `getTapId()`
- `loadBin()`
- `loadString()`
- `read()`
- `readBuf()`
- `readDap()`
- `resetCtrl()`
- `setDapAp()`
- `setSpeed()`
- `targetVoltage()`
- `write()`
- `writeBuf()`

- `writeDap()`

ServicesAPI

- `addService()`

TargetAPI

- `execute()`
- `getCpuName()`
- `getName()`
- `getState()`
- `halt()`
- `poll()`
- `readReg()`
- `reset()`
- `resume()`
- `writeReg()`

Error handling

- `catchString()`
- `catchThrowStr()`
- `errorToStr()`

Global functions

- `amregex()`
- `AnimateCursor()`
- `AnimateDone()`
- `astrsep()`
- `cliSwitch()`
- `FL_MAX()`
- `FL_MIN()`
- `Float2Integer()`
- `GetTickCount()`
- `IN_MAX()`
- `IN_MIN()`
- `include()`
- `Integer2Float()`
- `isInteger()`
- `isnull()`

- *isScriptObject()*
- *isString()*
- *require()*
- *Sleep()*
- *strToNumber()*
- *UN_MAX()*
- *UN_MIN()*

Logging

- *debugf(fmt, ...)*
- *errorf(fmt, ...)*
- *noticef(fmt, ...)*
- *printf(fmt, ...)*
- *tracef(fmt, ...)*
- *userf(fmt, ...)*
- *warnf(fmt, ...)*
- *yesNof(default, fmt, ...)*

Debugger

- *brkPt()*

CHAPTER

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